

Unit 22

ANALYTICAL TECHNIQUES

56 items · 151 marks

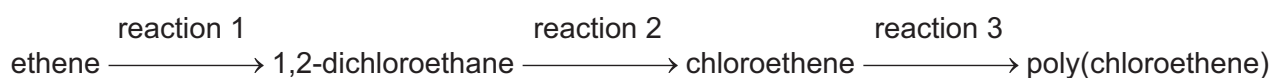
Syllabus sections in this unit

22.1 Infrared spectroscopy

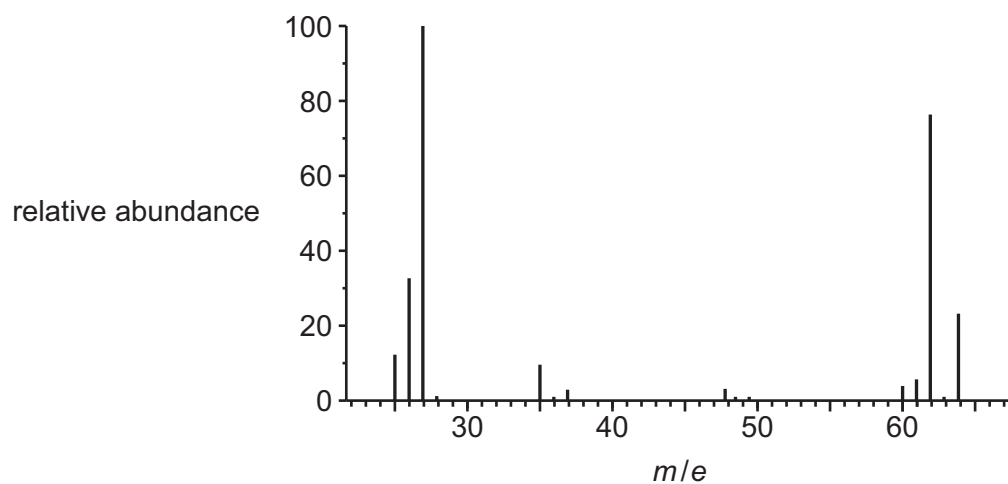
22.2 Mass spectrometry

Unit 22: Analytical techniques**1. M/J 26/P21/Q5/g**

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (g)** Figure 5.3 shows the mass spectrum of chloroethene.

**Figure 5.3**

- (i)** Identify the species responsible for the peaks at $m/e = 64$ and $m/e = 27$.

peak at $m/e = 64$

peak at $m/e = 27$

[2]

- (ii)** Explain the difference in relative abundance of the peaks at $m/e = 62$ and $m/e = 64$.

[2]

2. M/J 26/P22/Q6/c(i,ii)

6 (a) Figure 6.1 shows the structure of compound **P**.

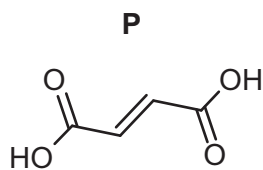


Figure 6.1

(c) An unlabelled sample contains one of compounds **P**, **R** or **S**.

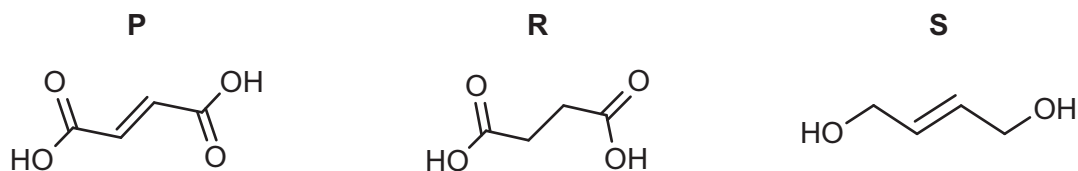


Figure 6.4

The mass spectrum of the sample is shown in Figure 6.5.

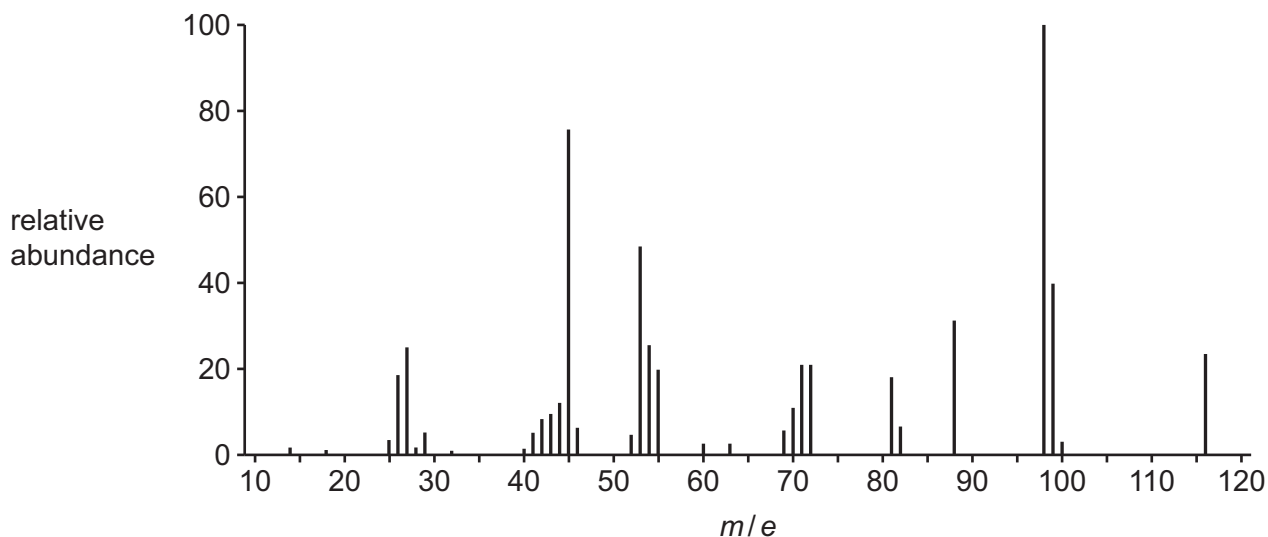


Figure 6.5

(i) Identify the **one** peak on Figure 6.5 which confirms that the sample is **P**.

Explain your reasoning.

[2]

(ii) Suggest the structural formulae of the fragments responsible for the peaks at $m/e = 45$ and $m/e = 99$.

structural formula of fragment at $m/e = 45$

structural formula of fragment at $m/e = 99$

[2]

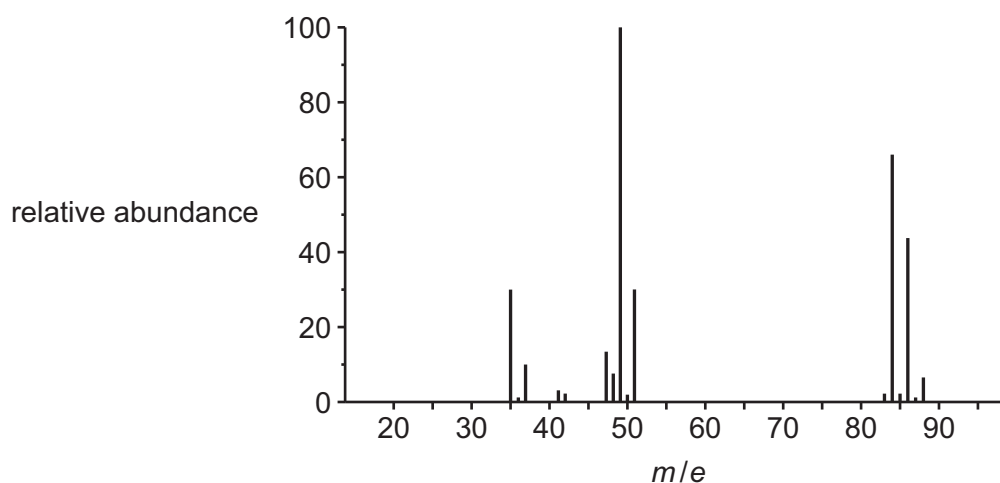
3. M/J 26/P23/Q5/c

- 5 Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

- (c) Compound **A** has the molecular formula CH_2Cl_2 . Figure 5.2 shows the mass spectrum of **A**.

**Figure 5.2**

- (i) Explain the difference in relative abundance of the peaks at $m/e = 35$ and $m/e = 37$.

[2]

- (ii) Complete Table 5.2.

Table 5.2

m/e value of peak	molecular formula of species responsible for peak
84	
86	
88	

[2]

4. M/J 26/P24/Q4/c

(c) Compound **W** is an aliphatic primary alcohol.

W is oxidised by acidified KMnO_4 when heated under reflux. Figure 4.2 shows the infrared spectrum of the resulting product.

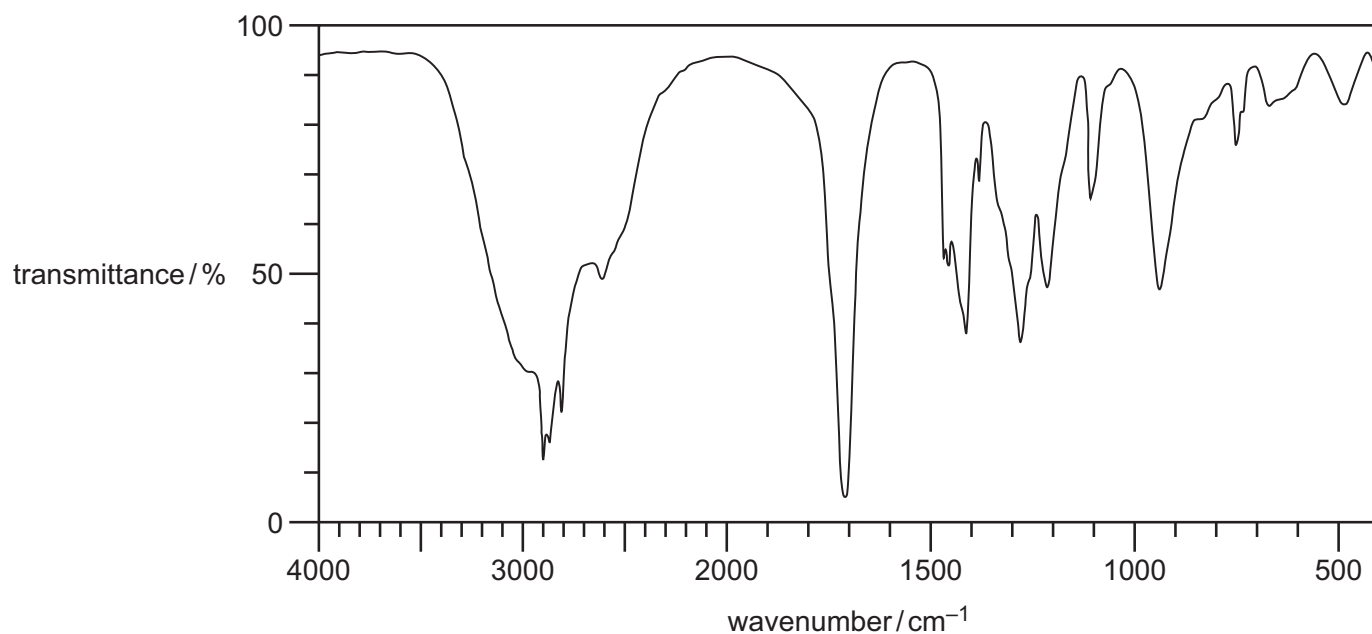


Figure 4.2

Table 4.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/ cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

The infrared spectrum in Figure 4.2 confirms that **W** is a primary alcohol.

Use Figure 4.2 to identify **two** absorbances that confirm **W** is a primary alcohol. State the bond and functional group containing the bond for each absorbance.

[2]

5. O/N 25/P21/Q3/e(ii)

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(e) Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

(ii) The infrared spectrum of **W** is shown in Fig. 3.3.

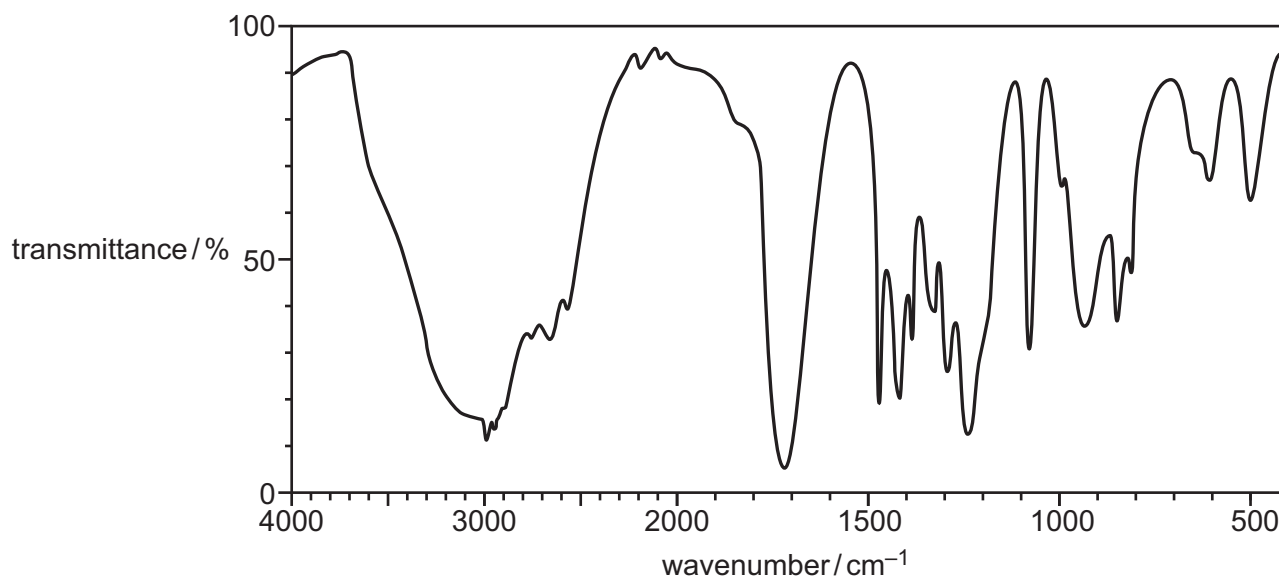


Fig. 3.3

Identify **two** absorptions in the infrared spectrum of **W** that would **not** be present in the infrared spectrum of cyclopentene.

- Write **1** or **2** on Fig. 3.3 against each of these two absorptions.
- Complete Table 3.2 to show which bond is responsible for each absorption that you have identified in Fig. 3.3.

Table 3.2

absorption	1	2
bond responsible		

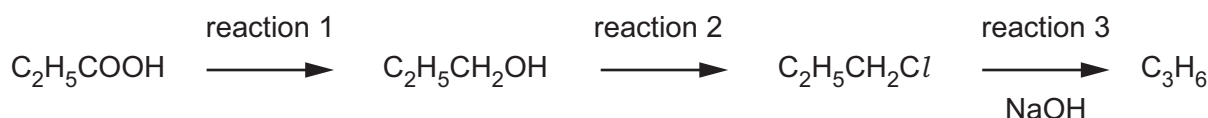
[2]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
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C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

6. O/N 25/P21/Q4/c(ii,iii)

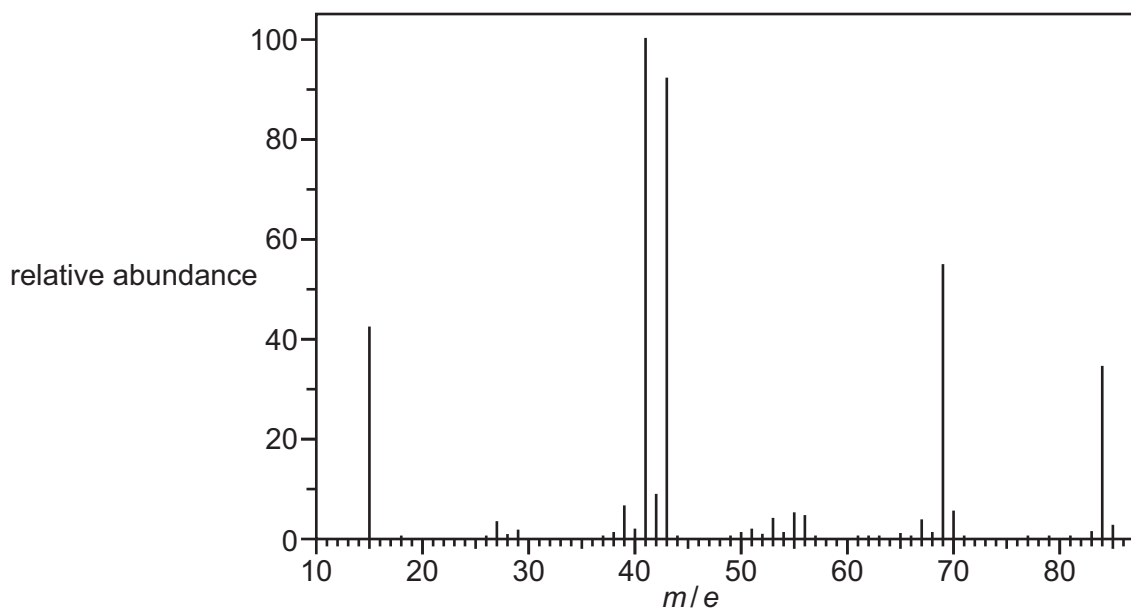
4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.



(ii) Fig. 4.2 shows the mass spectrum of either compound **X** or compound **Y**.

**Fig. 4.2**

Identify which of **X** and **Y** gives this mass spectrum.

Give **one** reason for your answer, referring to the fragmentation pattern.

[1]

(iii) The molecular ion peak in the spectrum in Fig. 4.2 has relative abundance 34.7.

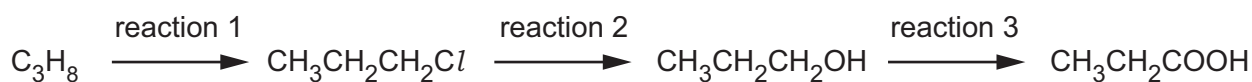
Calculate the relative abundance of the $[M+1]^+$ peak in this spectrum.

relative abundance of $[M+1]^+$ peak =

[1]

7. O/N 25/P22/Q4/d(ii,iii)

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

(d) CH_3CH_2COOH reacts with an unsaturated alcohol **R** to form unsaturated ester **S**.

(ii) The infrared spectrum of **S** is shown in Fig. 4.2.

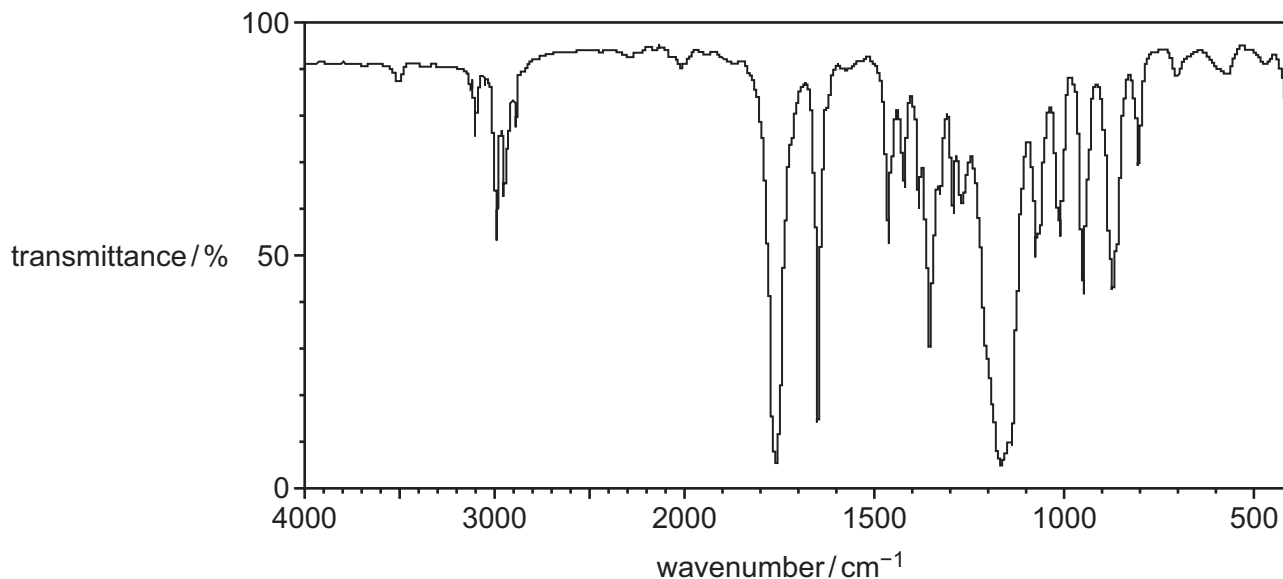


Fig. 4.2

Three absorptions in the infrared spectrum in Fig. 4.2 confirm that **S** is an ester and is unsaturated.

- Write **1**, **2** or **3** on Fig. 4.2 against each of these three absorptions.
- Complete Table 4.1 to show which bond is responsible for each absorption that you have identified in Fig. 4.2.

Table 4.1

absorption	1	2	3
bond responsible			

[3]

Table 4.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

(iii) The mass spectrum of **S** shows the following peaks.

Table 4.3

peak	relative abundance
M^+	4.7
$[M+1]^+$	0.31

Use Table 4.3 to calculate the number of carbon atoms in **S**.

number of carbon atoms in **S** =

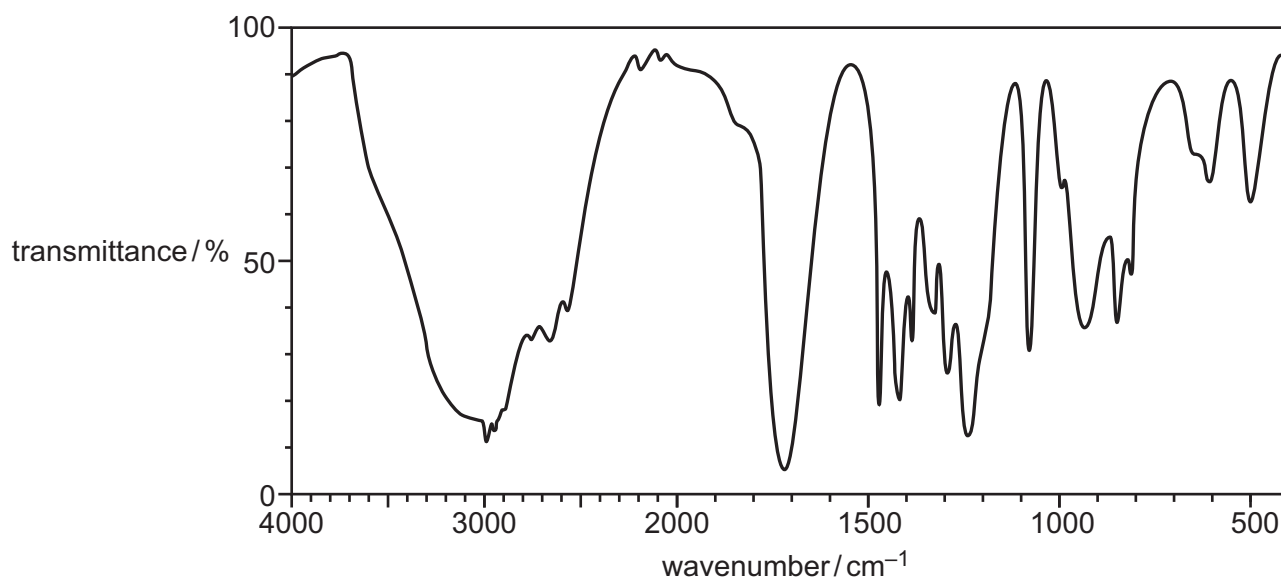
[1]

8. O/N 25/P23/Q3/e(ii)

- 3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(e) Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

(ii) The infrared spectrum of **W** is shown in Fig. 3.3.

**Fig. 3.3**

Identify **two** absorptions in the infrared spectrum of **W** that would **not** be present in the infrared spectrum of cyclopentene.

- Write **1** or **2** on Fig. 3.3 against each of these two absorptions.
- Complete Table 3.2 to show which bond is responsible for each absorption that you have identified in Fig. 3.3.

Table 3.2

absorption	1	2
bond responsible		

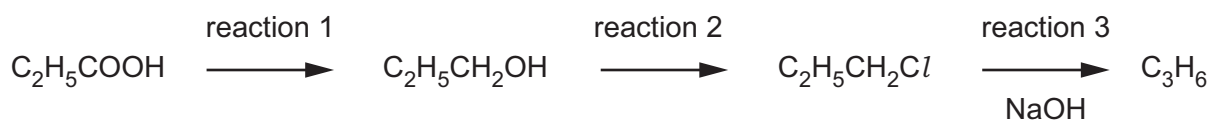
[2]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
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N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

9. O/N 25/P23/Q4/c(ii,iii)

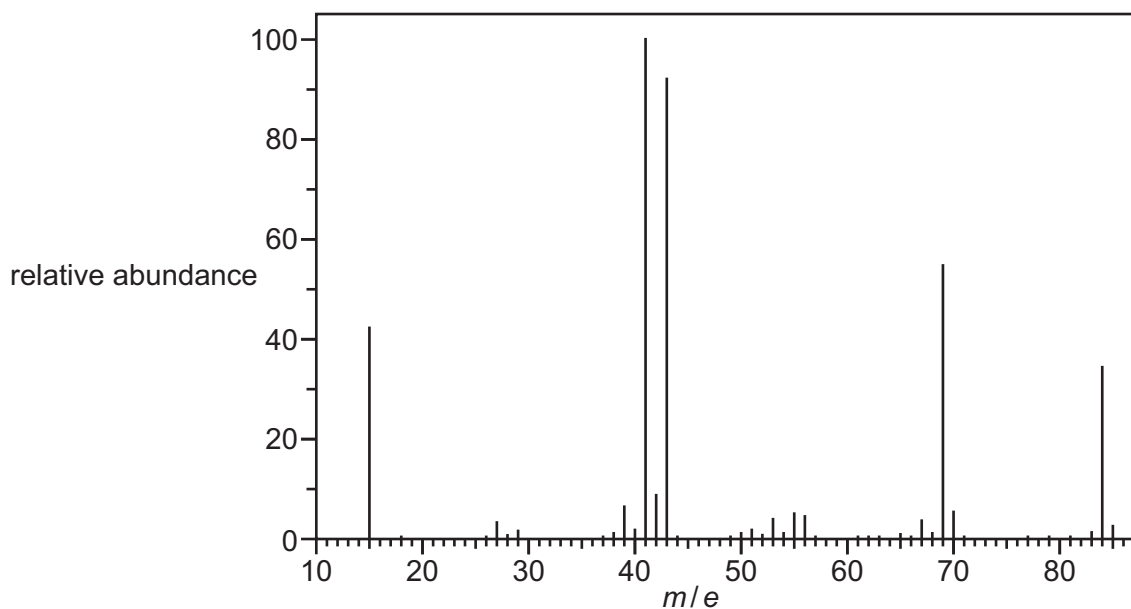
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**Fig. 4.1**

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.



(ii) Fig. 4.2 shows the mass spectrum of either compound **X** or compound **Y**.

**Fig. 4.2**

Identify which of **X** and **Y** gives this mass spectrum.

Give **one** reason for your answer, referring to the fragmentation pattern.

[1]

(iii) The molecular ion peak in the spectrum in Fig. 4.2 has relative abundance 34.7.

Calculate the relative abundance of the $[M+1]^+$ peak in this spectrum.

relative abundance of $[M+1]^+$ peak =

[1]

10. M/J 25/P21/Q4/c

4 The structure of vitamin C is shown in Fig. 4.1.

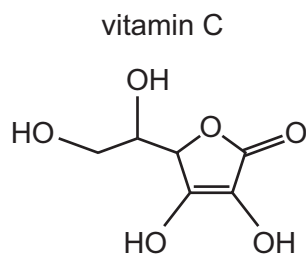


Fig. 4.1

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

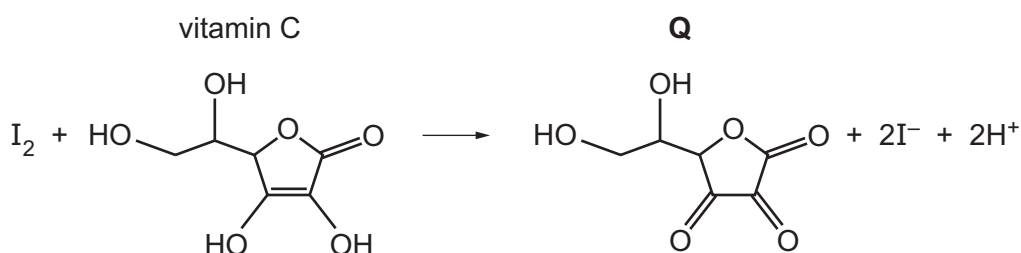


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm^3 of vitamin C solution. 5.00 cm^3 of this vitamin C solution is added to a flask with approximately 150 cm^3 of water and an indicator. Exactly 28.40 cm^3 of $5.00 \times 10^{-4}\text{ mol dm}^{-3}\text{ I}_2(\text{aq})$ reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(c) Predict **two** absorptions that will be seen in the infrared spectra of both vitamin C and **Q**. Describe the relevant bond and the specific functional group that is responsible for each absorption identified.

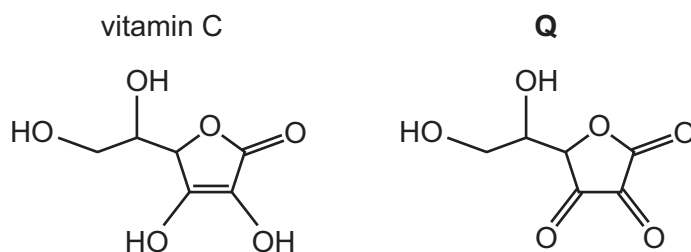
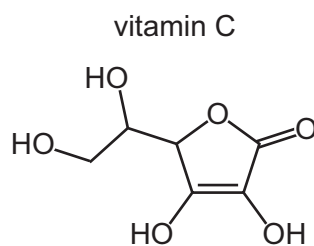


Fig. 4.3

[2]

11. M/J 25/P22/Q5/b(iii)

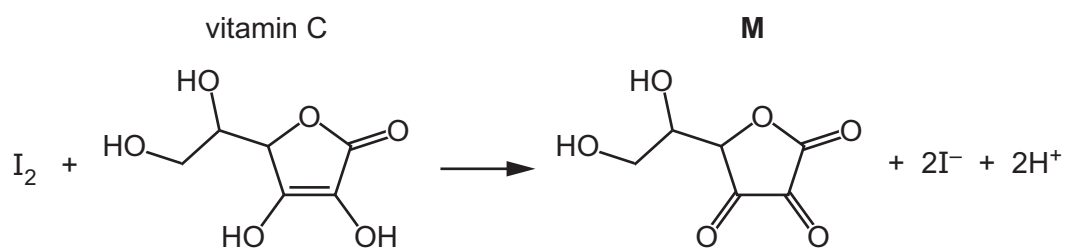
5 The structure of vitamin C is shown in Fig. 5.1.



(b) The mass of vitamin C present in 150.0 g of lemon is found in an experiment.

stage 1 All the vitamin C in 150.0g of lemon is extracted and dissolved in water to make 100.0 cm³ of solution L.

stage 2 A 25.0 cm³ sample of solution L is titrated with 5.00 × 10⁻³ mol dm⁻³ I₂(aq).



Exactly 36.65 cm³ of I₂(aq) reacts with the 25.0 cm³ sample of solution L.

[M_r : vitamin C, 176]

(iii) The progress of the reaction of vitamin C with $I_2(aq)$ to produce **M** is monitored using infrared spectroscopy.

Table 5.1 indicates the presence of some absorptions in the infrared spectrum of vitamin C. Complete Table 5.1 to predict which of these absorptions, if any, are present in the infrared spectrum of **M**.

Table 5.1

absorption/cm ⁻¹	present in spectrum of vitamin C	present in spectrum of M
1500–1680	✓	
2850–2950	✓	
3200–3650	✓	

[1]

Table 5.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

12. M/J 25/P23/Q5/b(v)

5 (a) **W** is a colourless liquid.

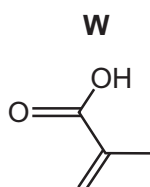


Fig. 5.1

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.



A
B
C

D

[2]

Table 5.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

13. M/J 25/P24/Q6/c

6 **P**, **Q** and **R** are three different hydrocarbon molecules.

(c) The mass spectrum of hydrocarbon **R** is recorded.

Information about the two peaks with m/e greater than 135 is shown in Fig. 6.1.

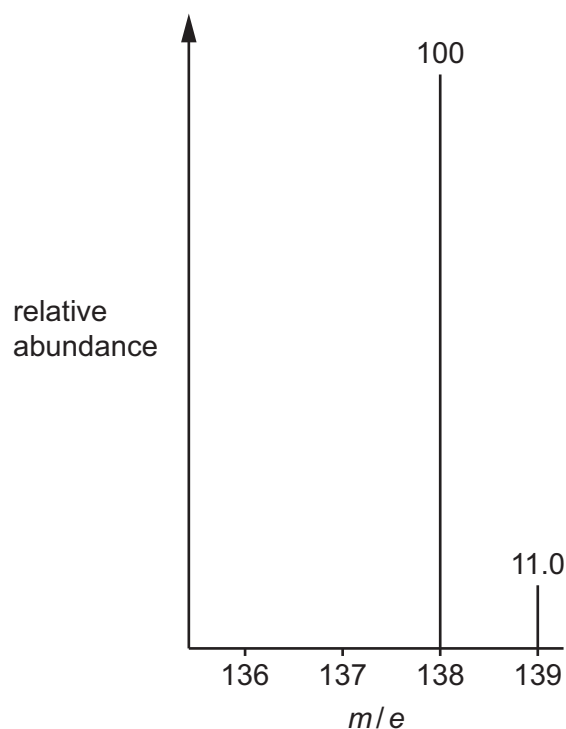


Fig. 6.1

(i) Use Fig. 6.1 to deduce the number of carbon atoms in a molecule of **R**.

Show your working.

number of carbon atoms = [2]

(ii) Use Fig. 6.1 and your answer to **(c)(i)** to deduce the molecular formula of **R**.

[1]

(iii) Suggest the molecular formula of the fragment of **R** with $m/e = 57$.

[1]

14. MAR 25/P22/Q1/d

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(d) Compound **D** contains carbon, hydrogen and chlorine **only**.

Fig. 1.2 shows the mass spectrum of **D**.

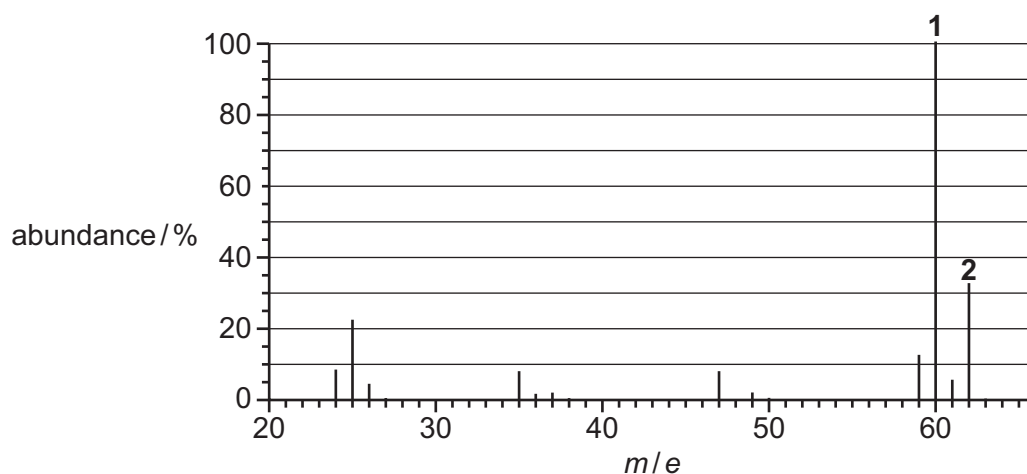


Fig. 1.2

(i) Explain the relative abundance of peaks **1** and **2** in the mass spectrum in Fig. 1.2.

[1]

(ii) Suggest the structure of **D**.

[1]

15. O/N 24/P21/Q4/f

4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

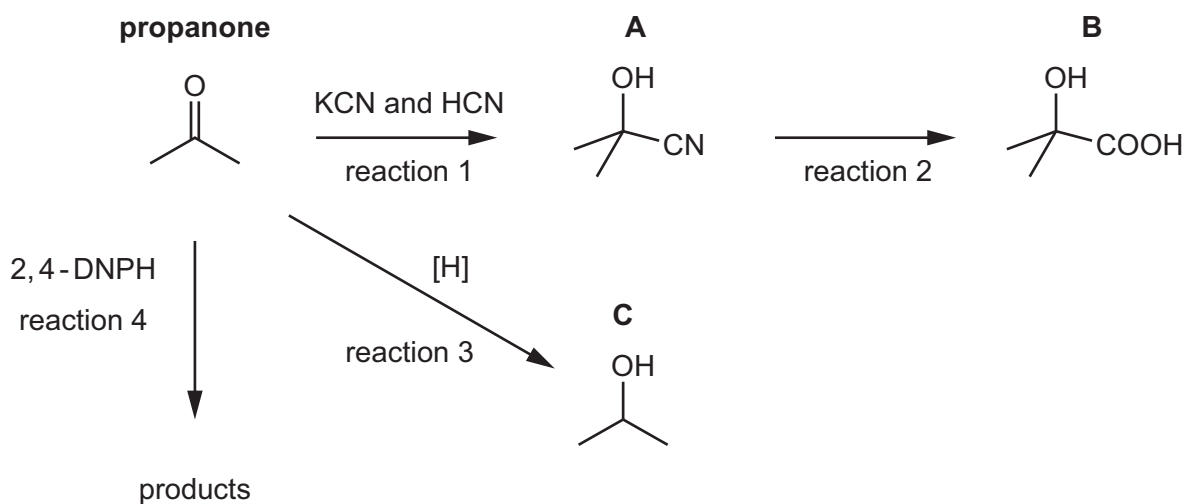


Fig. 4.1

(f) Compounds **A**, **B** and **C** can be distinguished using infrared spectroscopy.

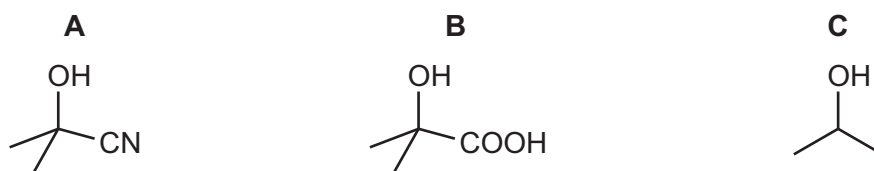


Fig. 4.3 shows the infrared spectrum of one of the compounds.

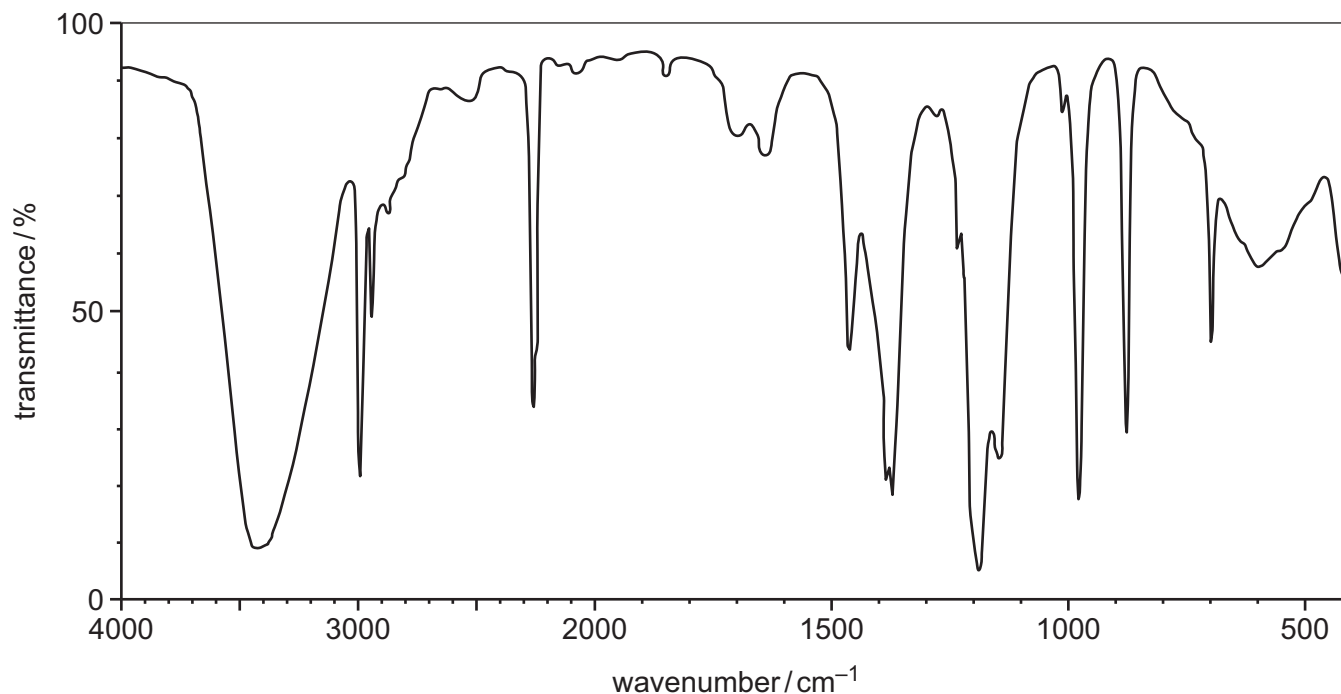


Fig. 4.3

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

- (i) Explain why the absorptions at 2850–2950 cm⁻¹ are **not** useful to help determine which of the compounds **A**, **B** or **C** produces the infrared spectrum in Fig. 4.3.

Use Table 4.1 to answer this question.

[1]

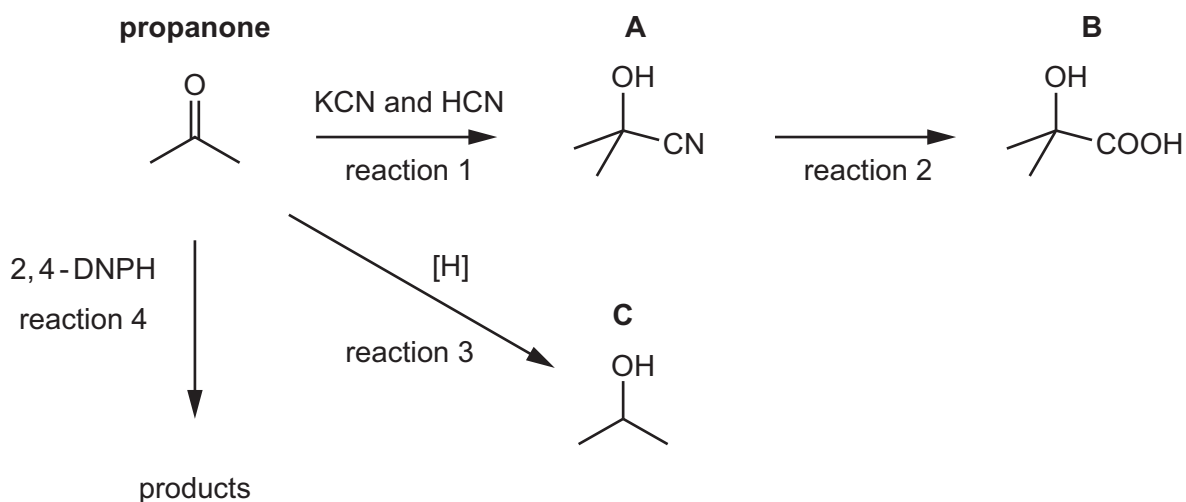
- (ii) Identify which of compounds **A**, **B** or **C** produces the infrared spectrum in Fig. 4.3. Explain your answer.

compound
explanation

[1]

16. O/N 24/P23/Q4/f

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (f) Compounds **A**, **B** and **C** can be distinguished using infrared spectroscopy.

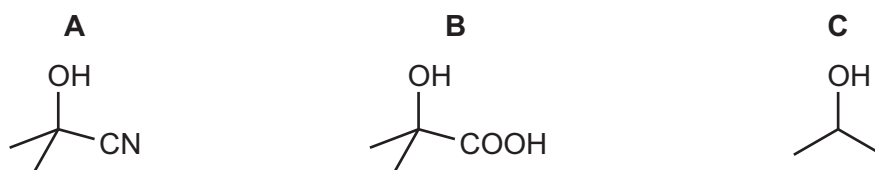


Fig. 4.3 shows the infrared spectrum of one of the compounds.

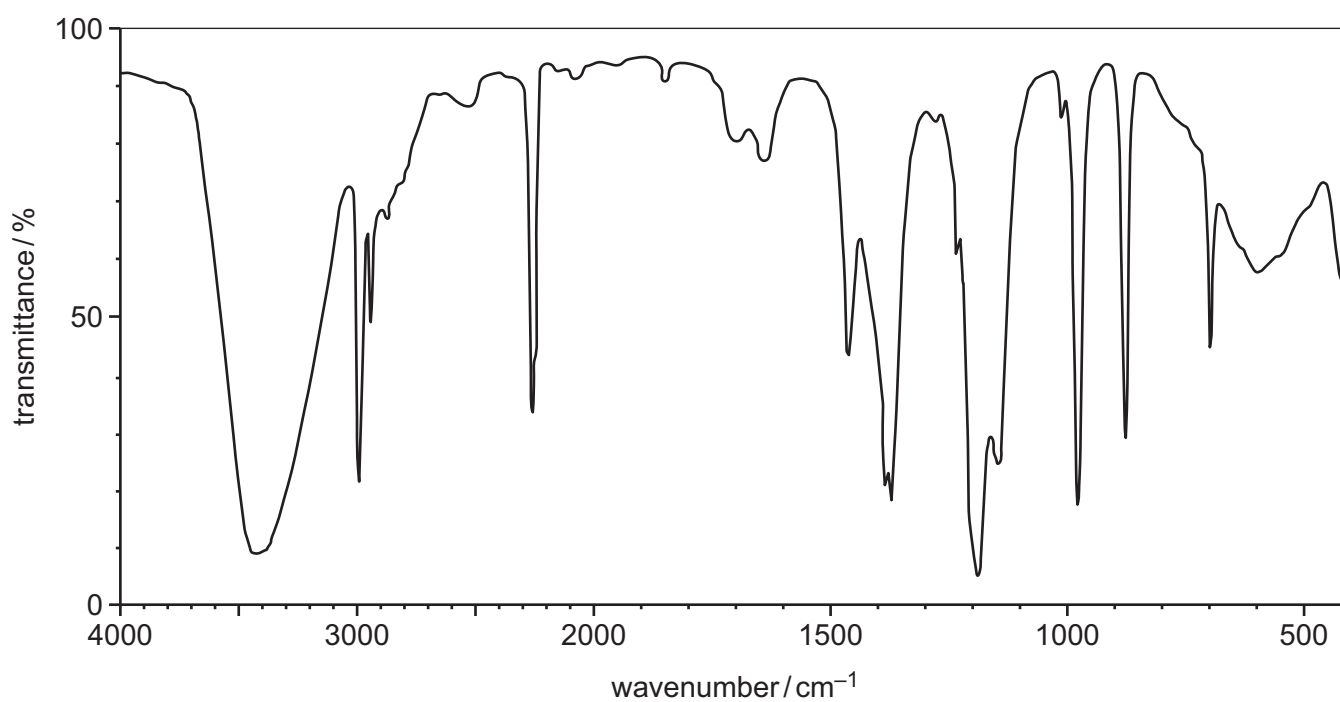
**Fig. 4.3**

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
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C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

- (i) Explain why the absorptions at 2850–2950 cm⁻¹ are **not** useful to help determine which of the compounds **A**, **B** or **C** produces the infrared spectrum in Fig. 4.3.

Use Table 4.1 to answer this question.

[1]

- (ii) Identify which of compounds **A**, **B** or **C** produces the infrared spectrum in Fig. 4.3. Explain your answer.

compound

explanation

[1]

17. M/J 24/P21/Q6/c(i,iv)

- (c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH₃OH, as shown in Fig. 6.3.

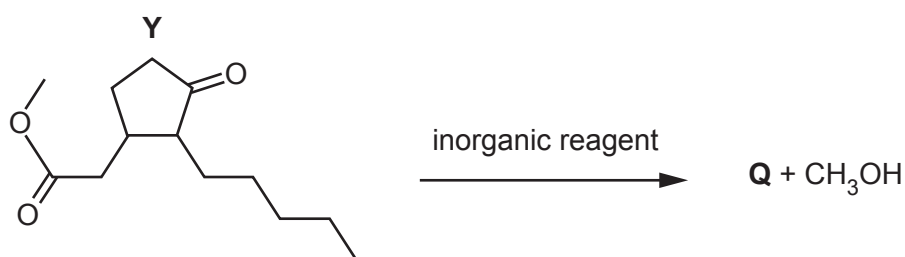


Fig. 6.3

- (i) Part of the mass spectrum for **Q** is shown in Fig. 6.4. Only peaks with m/e greater than 198 are shown.

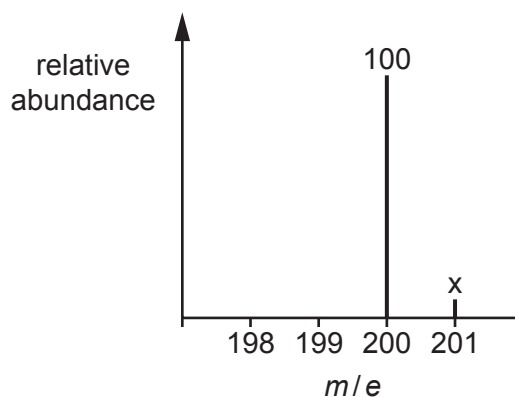


Fig. 6.4

Calculate the relative abundance, x , of the peak at $m/e = 201$.

Show your working.

$x =$ [2]

- (iv) Use Table 6.2 to describe and explain **two** differences between the infrared spectrum of **Y** and **Q** in the region above 1500 cm^{-1} .

[2]

Table 6.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

18. M/J 24/P23/Q5/c

- 5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

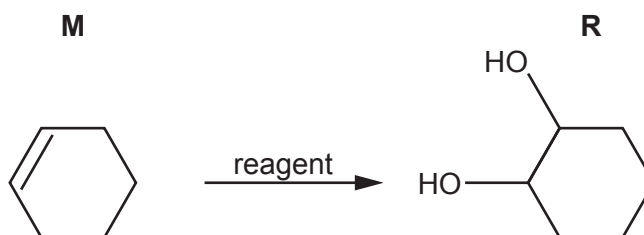


Fig. 5.1

Identify the reagent and conditions for this reaction.

[1]

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

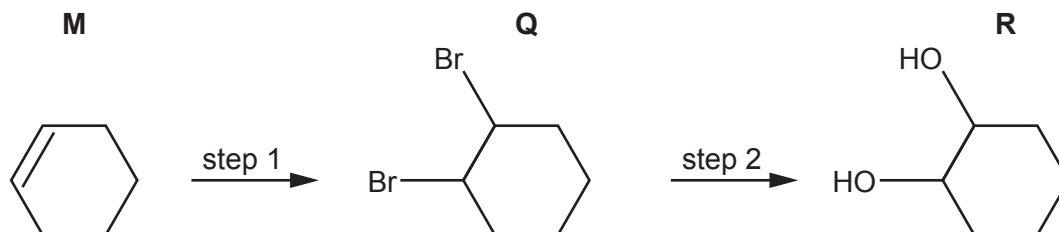


Fig. 5.2

(c) The infrared spectrum of **R** is shown in Fig. 5.3.

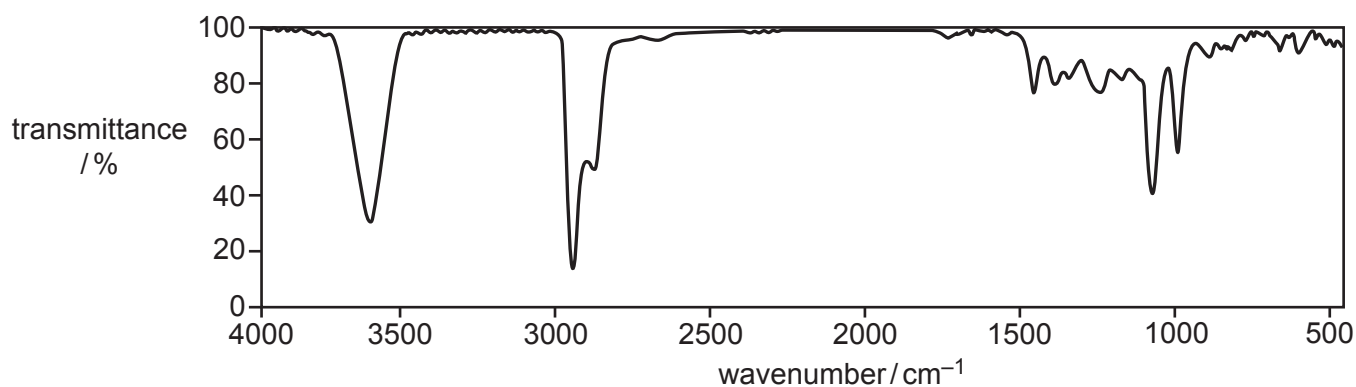


Fig. 5.3

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C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Use the absorptions in the region above 1500 cm⁻¹ in Table 5.1 when answering this question.

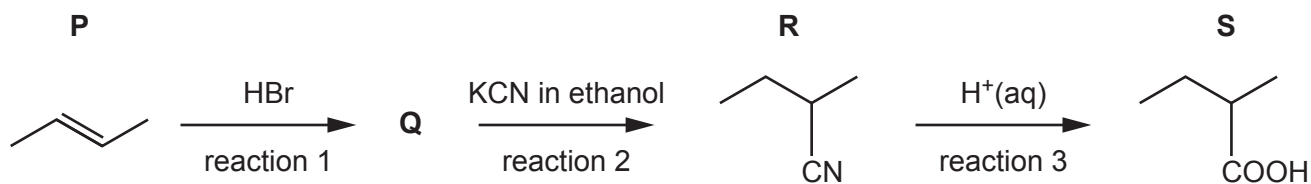
- Add **F** to Fig. 5.3 to identify the peak that is present in an infrared spectrum of both **Q** and **R**. Identify the bond that corresponds to the absorption for **F**.
- Add **G** to Fig. 5.3 to identify the peak that is **not** present in an infrared spectrum of **Q**. Identify the bond that corresponds to the absorption for **G**.

[2]

19. MAR 24/P22/Q4/c

also Unit 18 (2 marks); this unit 4 marks

- 4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

- (c) Compounds **S** and **T** react to form organic compound **U**, which has a single functional group.



Table 4.2 shows some data from the mass spectrum of **U**.

Table 4.2

peak	relative abundance
M^+	7.2
$[\text{M}+1]^+$	0.55

- (i) Use the data from Table 4.2 to show that **U** contains 7 carbon atoms.

Show your working.

[2]

- (ii) Fig. 4.2 shows the infrared spectrum of **U**.

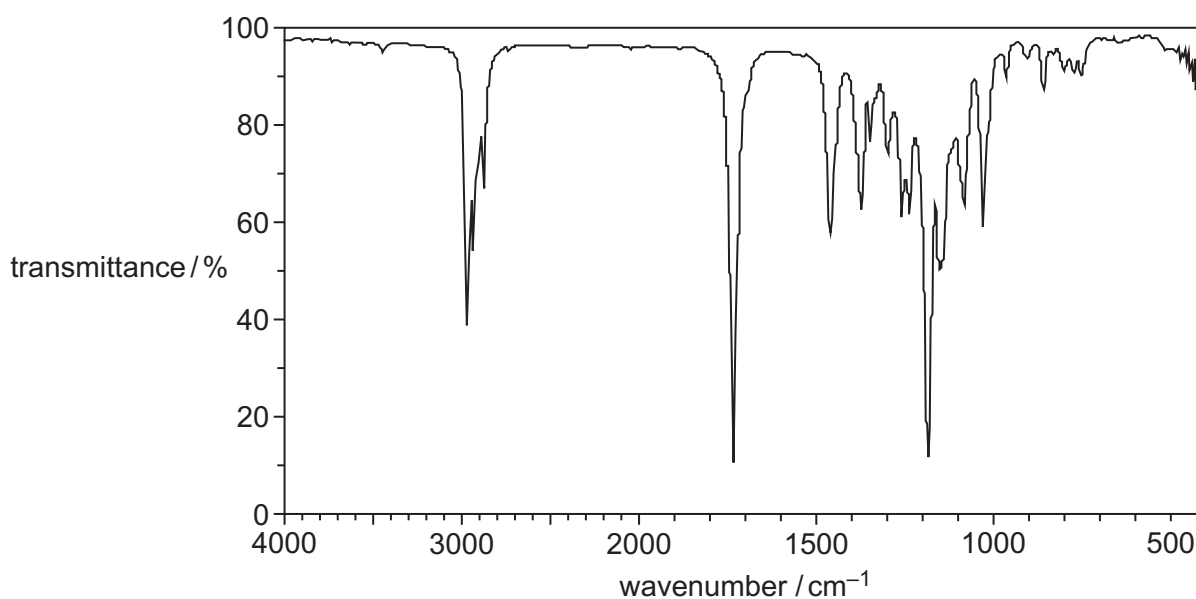
**Fig. 4.2**

Table 4.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Use Fig. 4.2 and Table 4.3 to identify the functional group present in **U**.

Explain your answer fully.

functional group

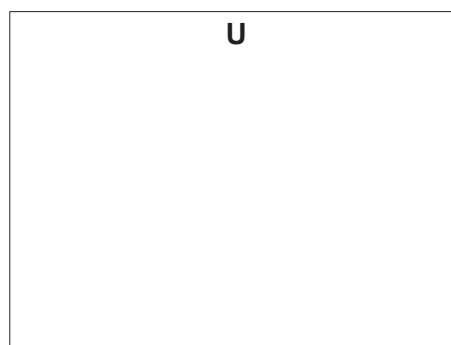
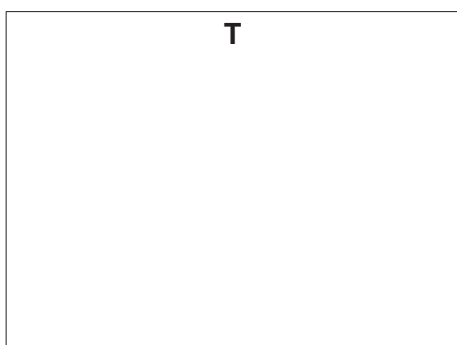
explanation

[2]

(iii) **T** also has a single functional group.

Use the information in (c)(i) and your answer to (c)(ii) to identify **T** and **U**.

Draw the structures of **T** and **U** in the boxes.



[2]

20. O/N 23/P21/Q4/b

- 4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10} .

**Fig. 4.1**

- (b) The mass spectrum of **C** shows a molecular ion peak at $m/e = 70$. This peak has a relative intensity of 48.7.

The relative intensity of the $[M+1]$ peak is 2.7.

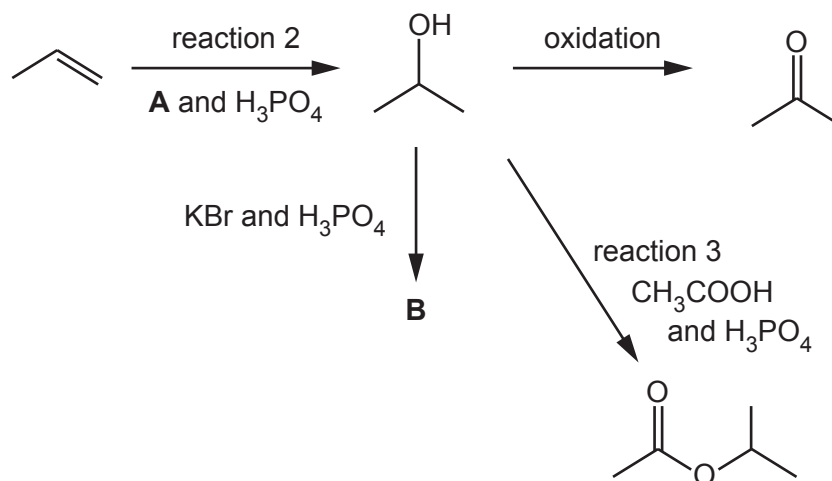
Show that this information is consistent with the molecular formula of **C**.

[2]

21. O/N 23/P22/Q3/c(iv),d(ii)

- 3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

- (c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

**Fig. 3.1**

- (iv) Reaction 3 is monitored using infrared spectroscopy. It is not possible to use the O—H absorption frequency to monitor the reaction.

Use Table 3.2 to identify a suitable bond whose absorption frequency can be used to monitor the progress of reaction 3.

State the change you would see in the infrared spectrum during reaction 3.

bond

change in infrared spectrum

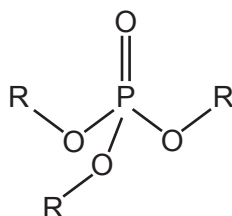
[2]

Table 3.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C–H	alkane	2850–2950

(d) H_3PO_4 also reacts with alcohols to form organophosphates.

Organophosphates are compounds similar to esters. They have the general structure shown in Fig. 3.2.



R = alkyl group

Fig. 3.2

(ii) Compound **T** is a simple organophosphate.

The mass spectrum of **T** shows a molecular ion peak at $m/e = 182$. This peak has a relative intensity of 12.7.

The relative intensity of the $M+1$ peak is 0.84.

Deduce the number of carbon atoms in **T**.
Hence suggest the molecular formula of **T**.

Assume that phosphorus and oxygen exist as single isotopes.

Show your working.

number of carbon atoms in **T** =

molecular formula of **T** =

[3]

22. O/N 23/P23/Q4/b

- 4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10} .

**Fig. 4.1**

- (b) The mass spectrum of **C** shows a molecular ion peak at $m/e = 70$. This peak has a relative intensity of 48.7.

The relative intensity of the $[M+1]$ peak is 2.7.

Show that this information is consistent with the molecular formula of **C**.

[2]

23. M/J 23/P21/Q3/f

- 3 Chlorine is a very reactive element.

- (f) **X** is a product of the substitution reaction that occurs when $CHClF_2$ reacts with Br_2 .

There is only one naturally occurring isotope of fluorine, ^{19}F .

The mass spectrum of **X** shows molecular ion peaks at $m/e = 164$, 166 and 168.

Complete Table 3.3 to show **all** the molecular ions responsible for each peak.

Table 3.3

m/e	formulae of molecular ions
164	
166	
168	$(CF_2^{37}Cl^{81}Br)^+$

[2]

24. M/J 23/P22/Q5/d(iv)

- (c) **C** is a structural isomer of **A** and **B**.

C is straight chained and has two functional groups.

C shows only one type of stereoisomerism.

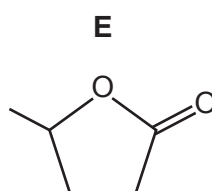
Table 5.2 describes observations when separate samples of **C** are added to different reagents.

Table 5.2

reagent	C
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	no reaction
$\text{Br}_2(\text{aq})$	orange to colourless
alkaline $\text{I}_2(\text{aq})$	yellow precipitate appears

(d) **D** reacts in the presence of a sulfuric acid catalyst to form **E** and water.

The structure of **E** is shown in Fig. 5.1.

**Fig. 5.1**

(iv) The infrared spectrum of **E** is shown in Fig. 5.2.

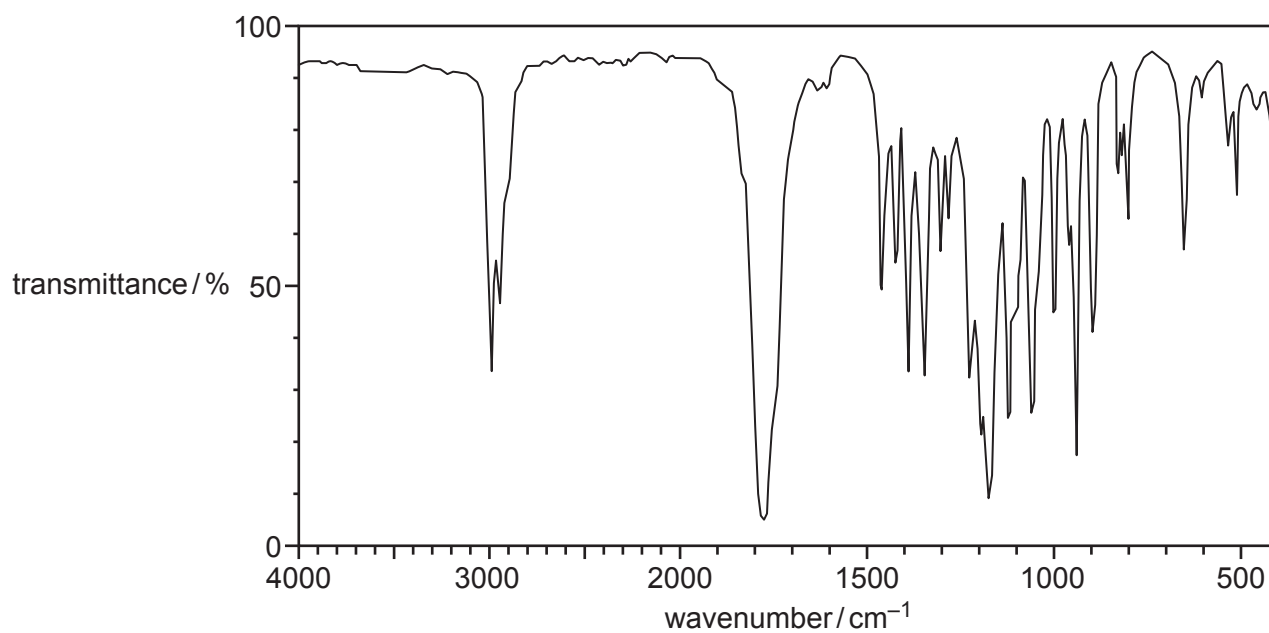
**Fig. 5.2**

Table 5.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

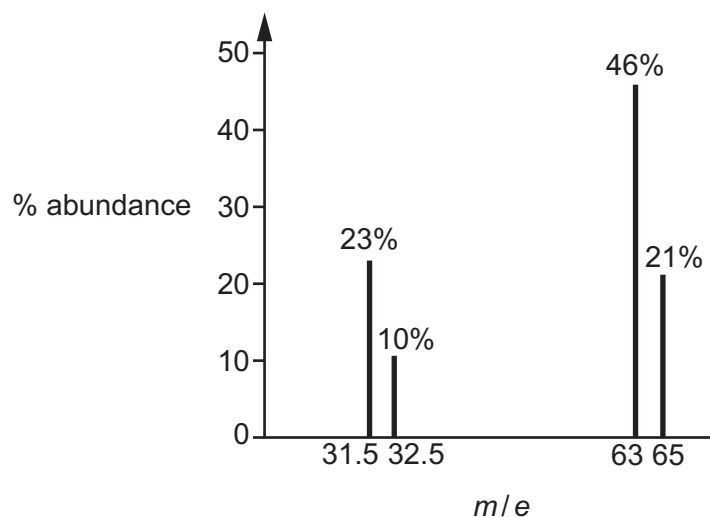
Use Fig. 5.2 and Table 5.3 to predict **two** differences in the absorptions above 1500 cm⁻¹ of the infrared spectrum of **D** compared to **E**. Explain your answer.

[2]

25. M/J 23/P23/Q1/c

- 1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

(c) The mass spectrum of a sample of pure copper is shown in Fig. 1.1.

**Fig. 1.1**

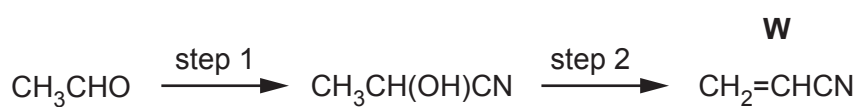
Identify the ion with an abundance of 23% in the sample.

[1]

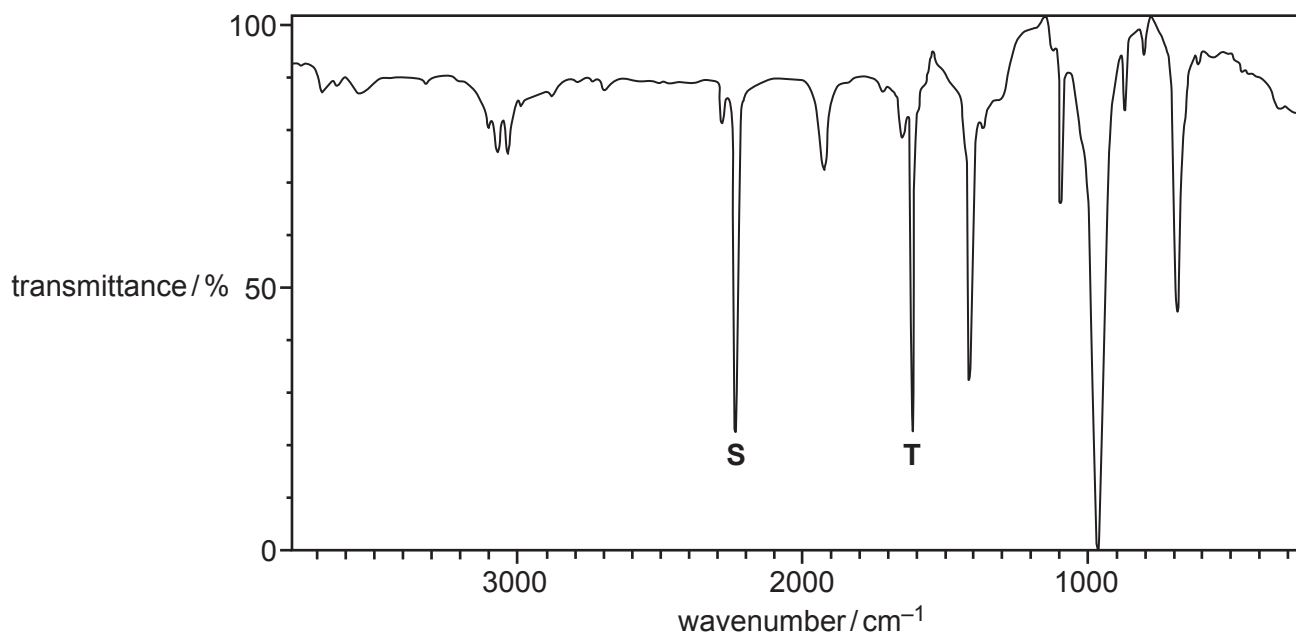
26. M/J 23/P23/Q6/b(iv)

- 6 Compound **W**, CH₂=CHCN, is used to make an addition polymer which is present in carbon fibres.

(b) CH₃CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH₃CHO is heated with HCN in the presence of KCN.

**Fig. 6.1**

(iv) Fig. 6.3 shows the infrared spectrum of **W**, $\text{CH}_2=\text{CHCN}$.

**Fig. 6.3****Table 6.1**

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

Use Table 6.1 to identify the bonds responsible for the absorptions marked **S** and **T** on Fig. 6.3.

S

T

27. MAR 23/P22/Q4/d

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

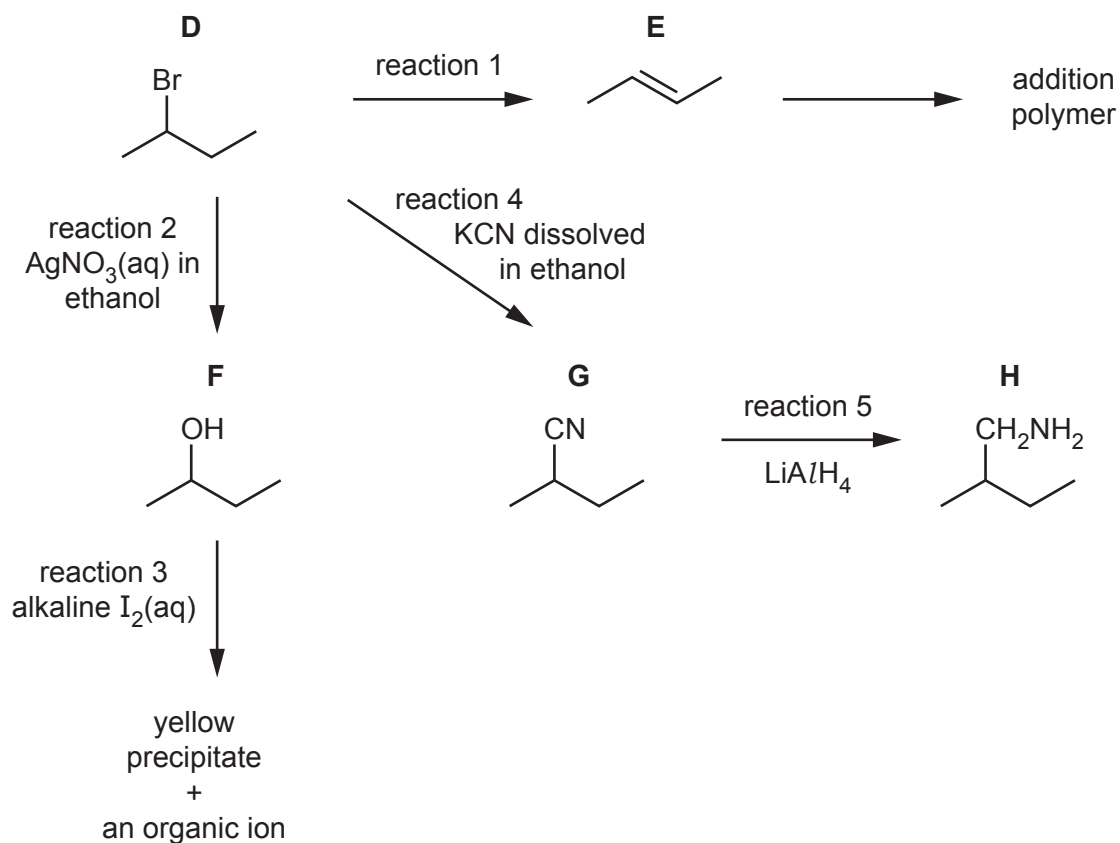


Fig. 4.1

(d) (i) Fig. 4.2 shows the infrared spectrum of one of the compounds **D**, **E**, **F**, **G** or **H**.

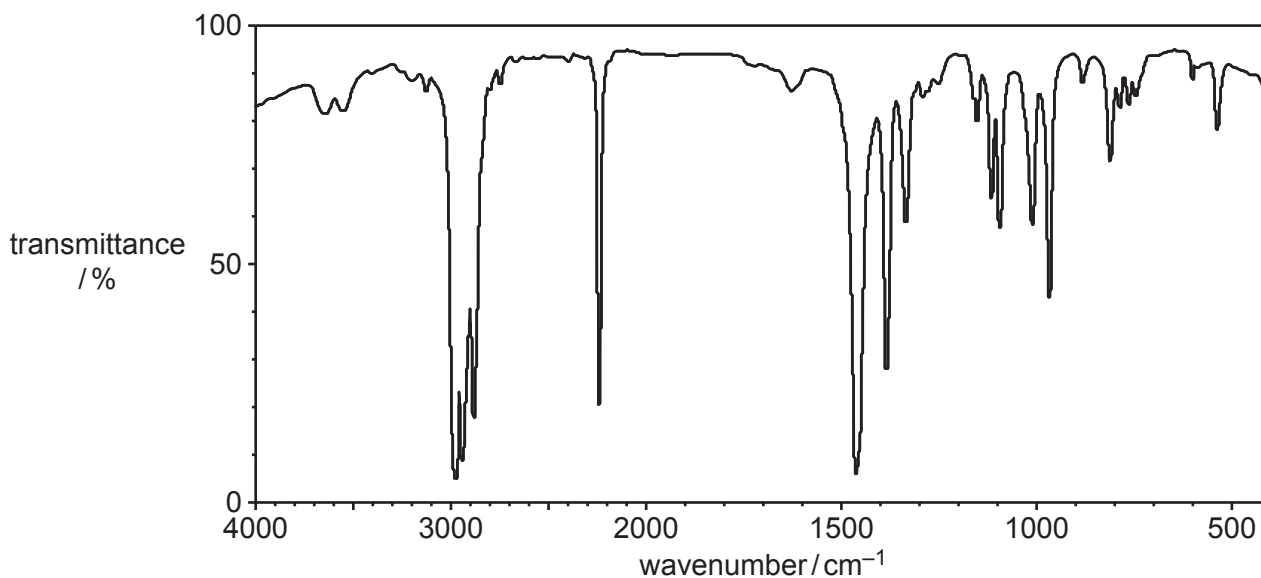


Fig. 4.2

Use information from Table 4.1 (on page 14) to identify which of the compounds **D**, **E**, **F**, **G** or **H** produces the infrared spectrum in Fig. 4.2.

Explain your answer.

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

(ii) In the mass spectrum of **D**, the relative abundance of the molecular ion peak is 3.4.

Predict the relative abundance of the M+2 peak for **D**.

Explain your answer.

[1]

28. O/N 22/P21/Q5/b,c

- 5 Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

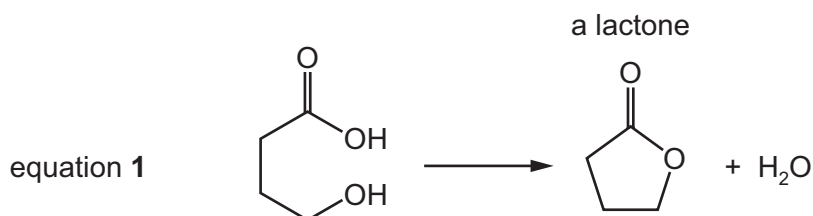


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

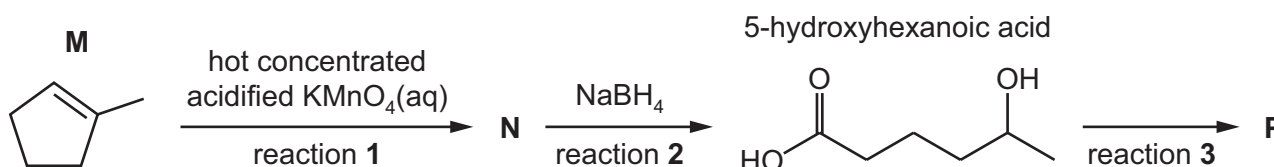


Fig. 5.1

(b) A student monitors the progress of reaction 2 using infrared spectroscopy.

Use Table 5.1 to suggest why it is difficult to distinguish between **N** and 5-hydroxyhexanoic acid using infrared spectroscopy.

[2]

Table 5.1

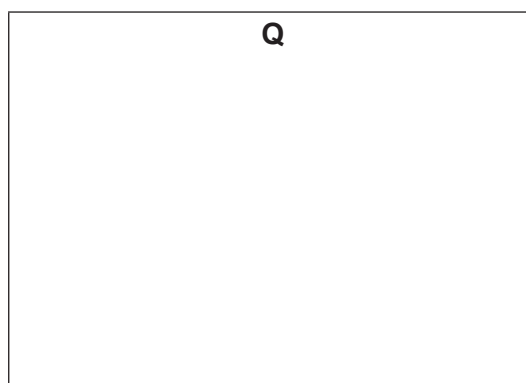
bond	functional group containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–3100
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

- (c) Unknown lactone **Q** is analysed using mass spectrometry.
Table 5.2 shows information from the mass spectrum.

Table 5.2

peak	<i>m/e</i>	abundance
M+	72	95.5
M+1	73	3.15

Use these data to deduce the structure of **Q**.
Show your working.



[2]

29. O/N 22/P22/Q3/c

also Unit 14 (1 mark); this unit 4 marks

- 3** Organic compounds can be distinguished using chemical tests and analytical techniques.

(c) Organic compound **E** contains three carbon atoms.

E reacts with cold dilute acidified $\text{KMnO}_4(\text{aq})$ to form a single compound **F** with $M_r = 154.9$.

Fig. 3.1 shows the infrared spectrum of **E**.

Fig. 3.2 shows the infrared spectrum of **F**.

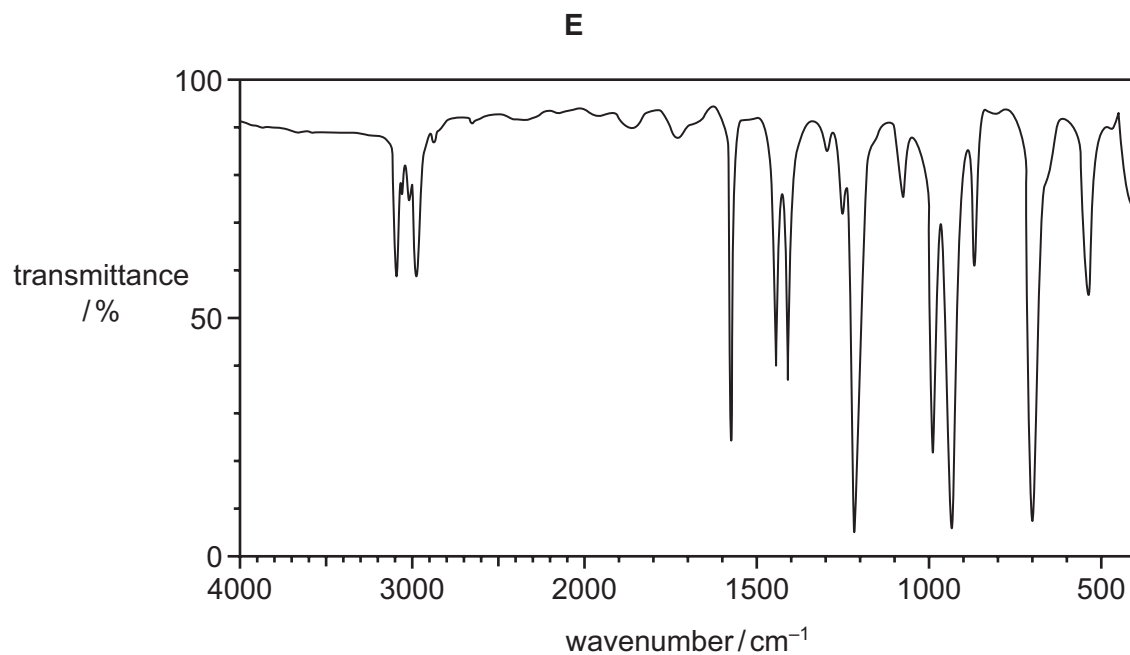


Fig. 3.1

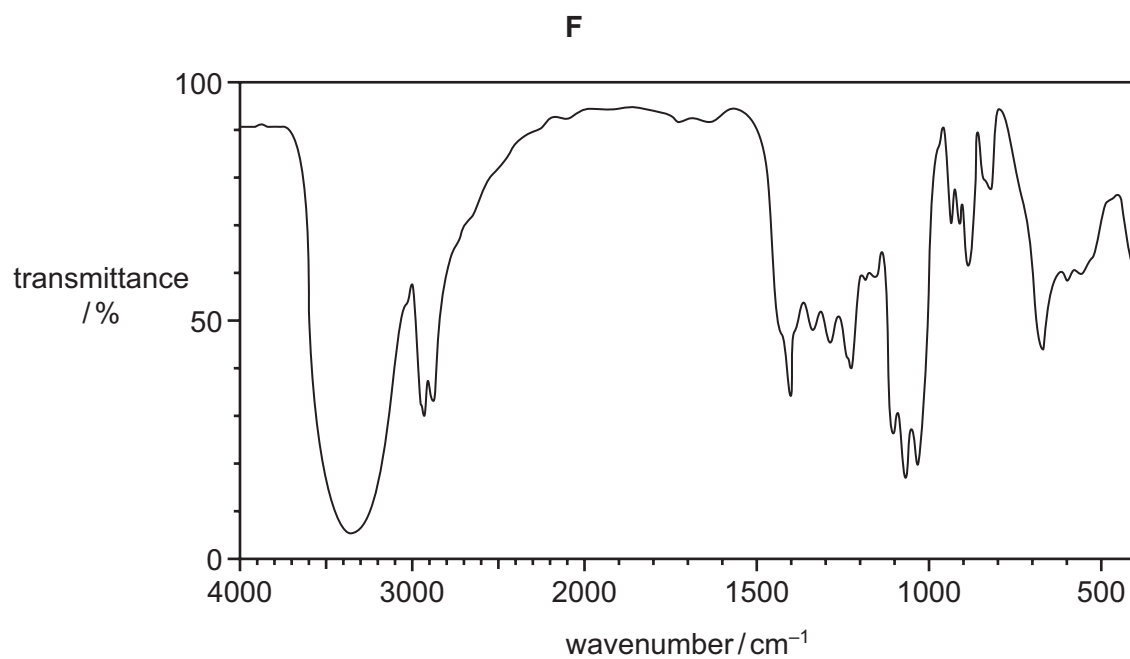


Fig. 3.2

Table 3.2

bond	functional group containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–3100
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Both spectra show absorptions between 2850 and 2950cm⁻¹ owing to C–H bonds in each molecule.

- (i) Use the two infrared spectra and Table 3.2 to identify the functional group present only in **E**.

Explain your answer, referring only to absorptions at frequencies greater than 1500cm⁻¹.

functional group

explanation

[1]

- (ii) Use the infrared spectrum of **F** to identify the functional group formed when **E** reacts with cold dilute acidified KMnO₄(aq).

Explain your answer, referring only to absorptions at frequencies greater than 1500cm⁻¹.

functional group

explanation

[1]

- (iii) The mass spectrum of **E** shows a molecular ion peak and an M+2 peak of approximately equal abundance at $m/e = 120$ and 122.

Deduce the relative molecular mass, M_r , of **E**.

$M_r =$ [1]

- (iv) Use the information in **3(c)** to suggest a structure for **E**.

[1]

- (v) Complete the equation for the reaction of **E** with cold dilute acidified KMnO₄(aq) to form **F**. In the equation, [O] represents cold dilute acidified KMnO₄(aq).



[1]

30. O/N 22/P23/Q5/b,c

- 5 Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.

Equation 1 shows an example of the formation of a lactone.

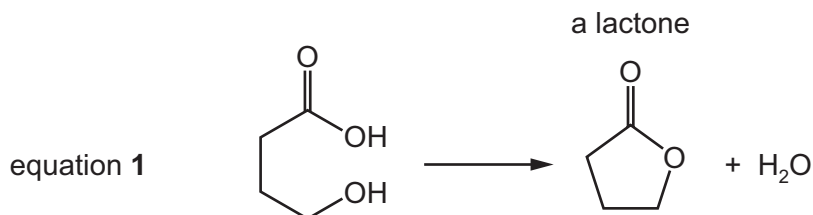


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

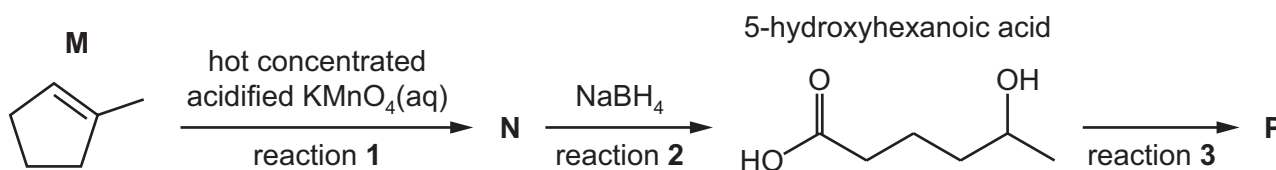


Fig. 5.1

- (b) A student monitors the progress of reaction 2 using infrared spectroscopy.

Use Table 5.1 to suggest why it is difficult to distinguish between **N** and 5-hydroxyhexanoic acid using infrared spectroscopy.

[2]

Table 5.1

bond	functional group containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–3100
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

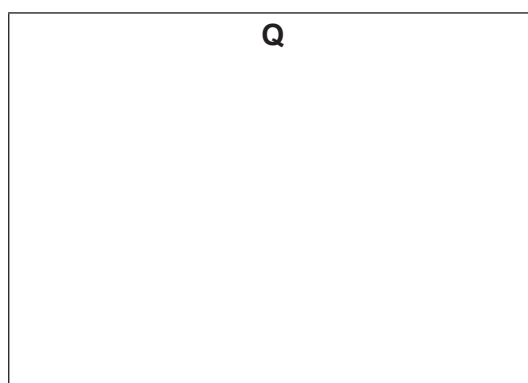
- (c) Unknown lactone **Q** is analysed using mass spectrometry.

Table 5.2 shows information from the mass spectrum.

Table 5.2

peak	<i>m/e</i>	abundance
M ⁺	72	95.5
M+1	73	3.15

Use these data to deduce the structure of **Q**.
Show your working.



[2]

31. M/J 22/P21/Q4/d

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

(d) Fig. 5.1 shows the mass spectrum of ketone **Z**, $\text{C}_5\text{H}_{10}\text{O}$.

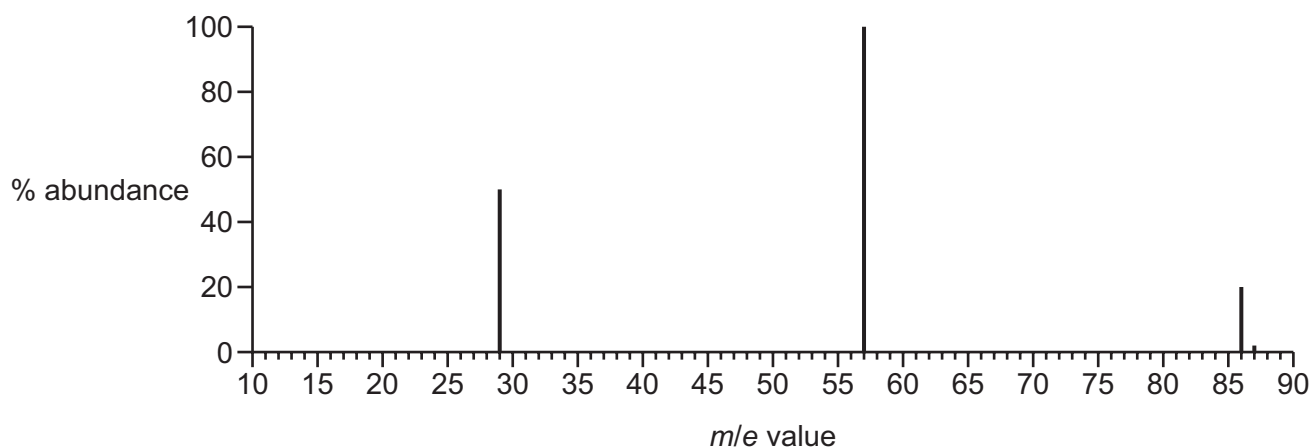


Fig. 5.1

Use the information in Fig. 5.1 to suggest the formulae of the fragments with m/e peaks at 29 and 57. Deduce the identity of **Z**.

$m/e = 29$

$m/e = 57$

identity of **Z**

[3]

32. M/J 22/P22/Q6/c(i,ii)

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

(c) Fig. 6.1 shows the mass spectrum of **Z**.

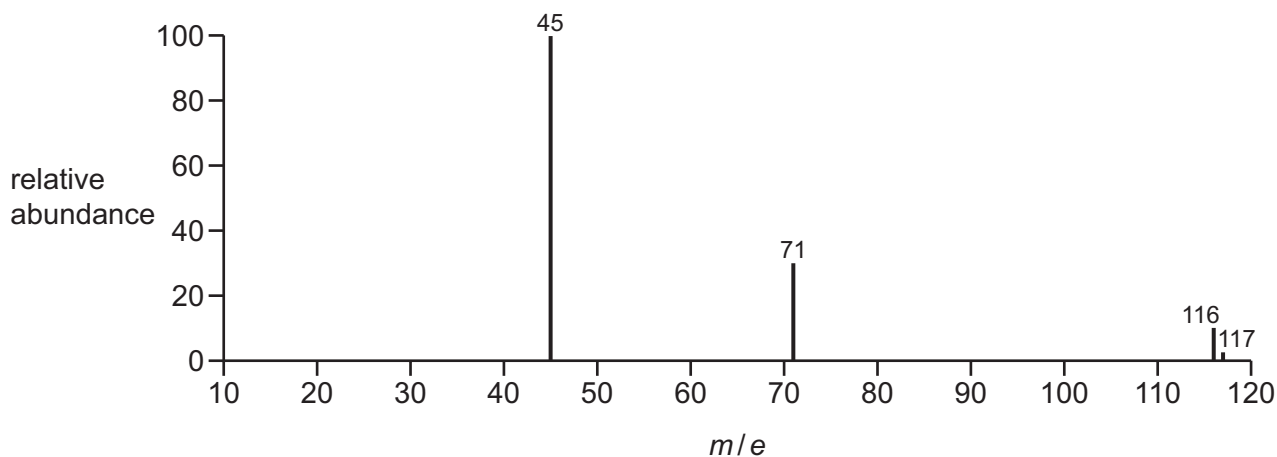


Fig. 6.1

- (i)** Deduce the molecular formula of **Z**. Explain your answer by referring to the molecular ion peak in Fig. 6.1 and the empirical formula of **Z**.

[1]

- (ii)** Use Fig. 6.1 to suggest the formulae of the fragments with m/e peaks at 45 and at 71.

m/e 45

m/e 71

[2]

33. M/J 22/P23/Q5/b

(b) **Z** contains three types of atom: carbon, hydrogen and a halogen. The mass spectrum of **Z** is recorded. Fig. 5.1 shows a section of the mass spectrum at m/e greater than 63.

The fragment at $m/e = 64$ is the molecular ion peak.

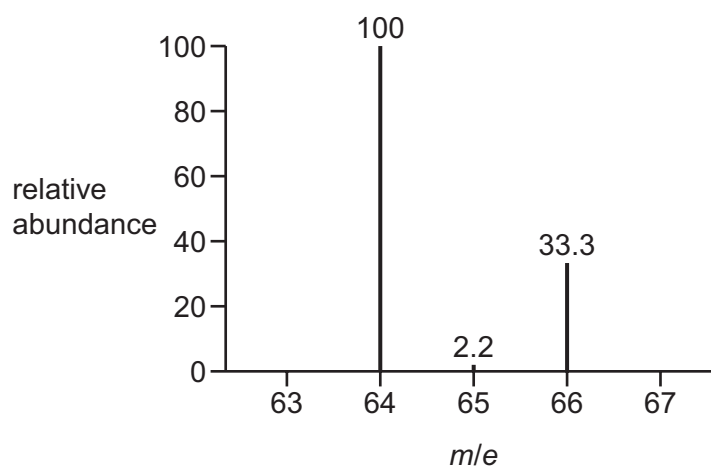


Fig. 5.1

- (i) Deduce the number of carbon atoms present in a molecule of **Z** using Fig. 5.1. Show your working.

[1]

- (ii) Deduce which halogen is present in **Z** using Fig. 5.1. Explain your answer.

[1]

- (iii) There are also peaks at $m/e = 29$ and $m/e = 49$.

Suggest the formulae of these fragments.
Deduce the name of **Z**.

$m/e = 29$

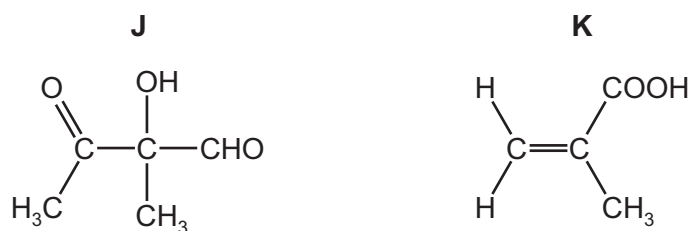
$m/e = 49$

name of **Z**

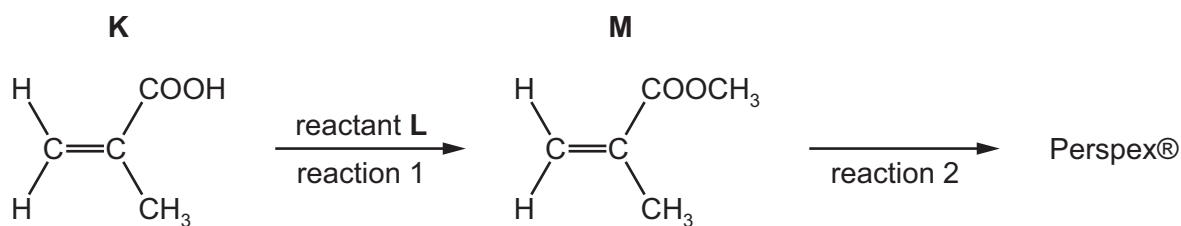
[3]

34. MAR 22/P22/Q4/b(iii)

4 Compounds **J** and **K** are found in plant oils.

**Fig. 4.1**

(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.

**Fig. 4.2**

(iii) Use information from Table 4.2 to suggest how the infrared spectra of **M** and Perspex® would differ. Explain your answer.

[1]

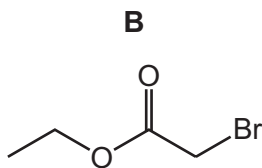
Table 4.2

bond	functional group containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–3100
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

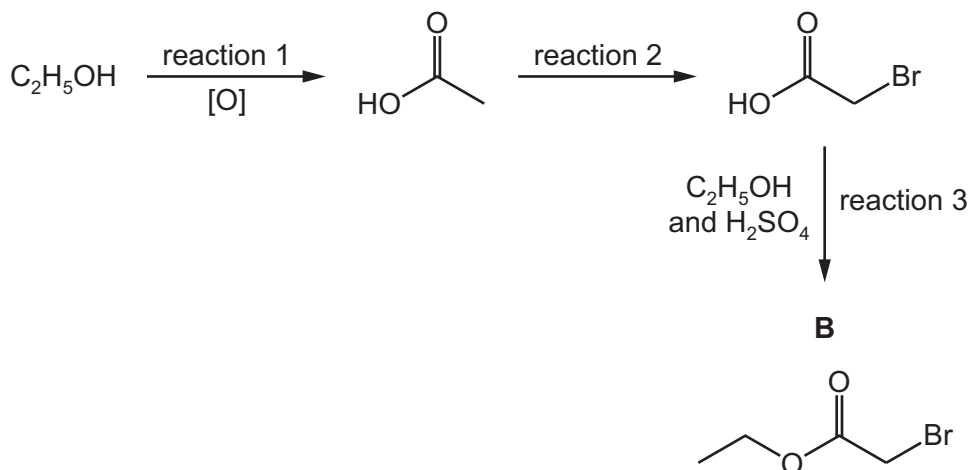
35. O/N 21/P21/Q4/b

Data Booklet needed

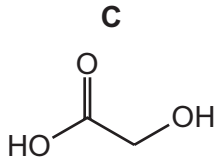
- 4** Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, $\text{C}_2\text{H}_5\text{OH}$.



- (b)** Reaction 2 needs to take place in the absence of water to prevent formation of compound **C**.



If **C** is present in the reaction mixture of reaction 3, a different compound, compound **D**, will also form. Compound **D** has two identical functional groups.

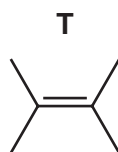
The infrared spectrum of **D** shows strong absorptions at 1100 cm^{-1} and 1720 cm^{-1} , but no absorption due to O–H bonds.

Use the *Data Booklet* to identify the functional group present in **D**.

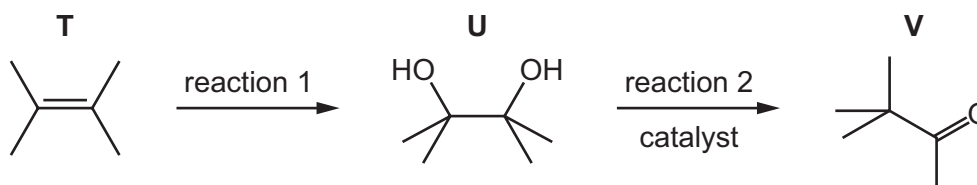
Explain your answer as fully as you can.

36. O/N 21/P22/Q3/d(iii)

3 Compound **T** is an isomer of C_6H_{12} .



(d) A reaction scheme starting with **T** is shown. Reaction 2 occurs in the presence of a catalyst; knowledge of the mechanism for this reaction is not required.



(iii) The progress of reaction 2 can be monitored by infrared spectroscopy.

The absorption caused by O–H bonds is always present because water is used as a solvent.

Identify two absorptions, and the bonds responsible for these absorptions, whose appearance will change significantly during the reaction.

1

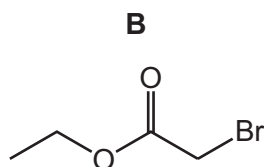
2

[2]

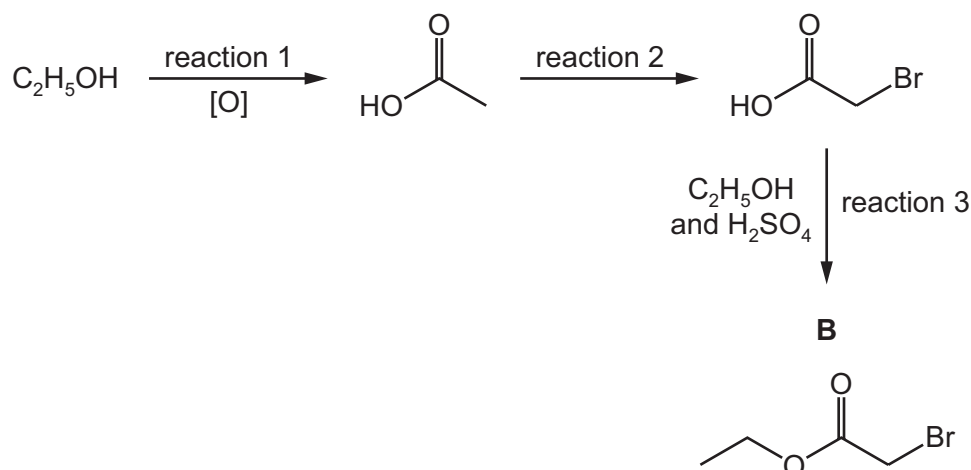
37. O/N 21/P23/Q4/b

Data Booklet needed

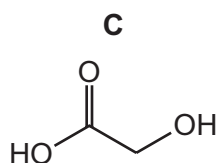
- 4** Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, C_2H_5OH .



- (b)** Reaction 2 needs to take place in the absence of water to prevent formation of compound **C**.



If **C** is present in the reaction mixture of reaction 3, a different compound, compound **D**, will also form. Compound **D** has two identical functional groups.

The infrared spectrum of **D** shows strong absorptions at 1100 cm^{-1} and 1720 cm^{-1} , but no absorption due to O–H bonds.

Use the *Data Booklet* to identify the functional group present in **D**.

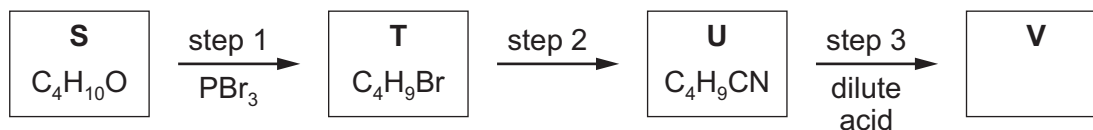
Explain your answer as fully as you can.

[3]

38. M/J 21/P22/Q5/b(v)

5 **S** is a secondary alcohol with molecular formula $C_4H_{10}O$.

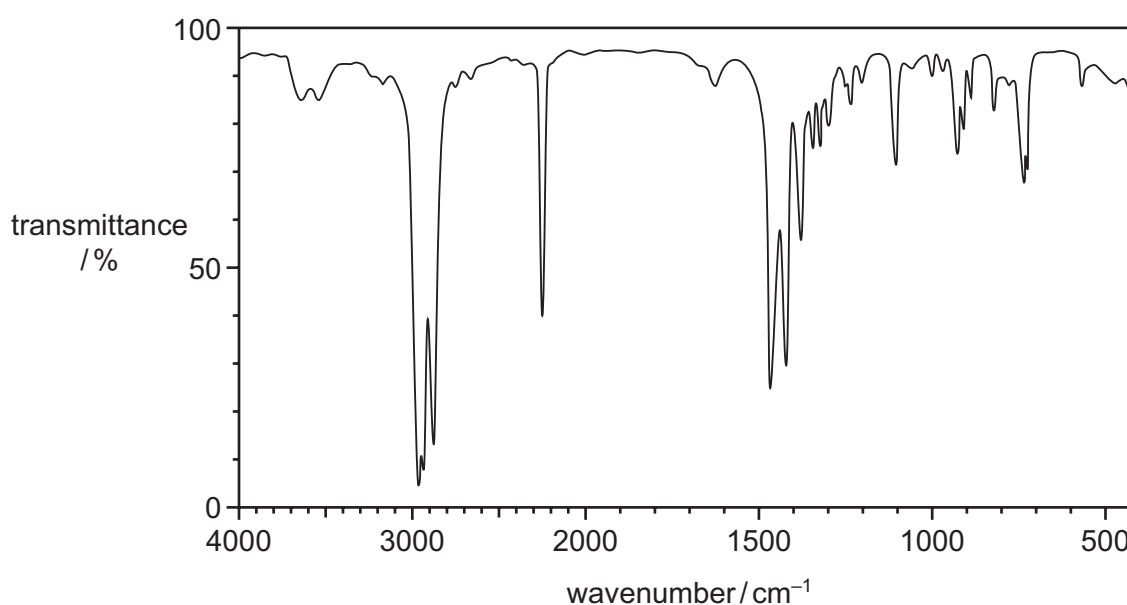
(b) **S** is converted to **V** in a three-step reaction sequence.



In step 1, the secondary alcohol **S** reacts with PBr_3 to produce **T**, which has molecular formula C_4H_9Br .

(v) An unlabelled sample contains either **S**, **T** or **U**.

The sample produces the infrared spectrum shown.

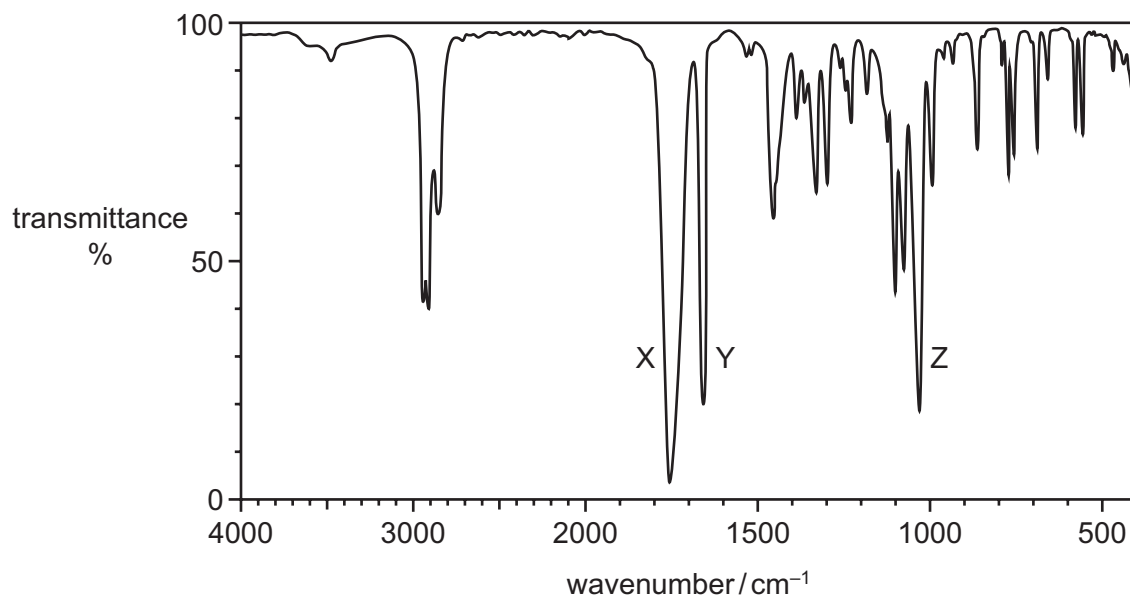


Explain how this spectrum confirms that the unknown sample contains **U**.

In your answer identify **one** relevant absorption in the infrared spectrum and the bond that corresponds to this absorption in the region above 1500 cm^{-1} .

[1]

The student analysed **L** using infrared spectroscopy. The following spectrum was obtained.



(i) Identify the bonds responsible for the absorptions marked X and Z.

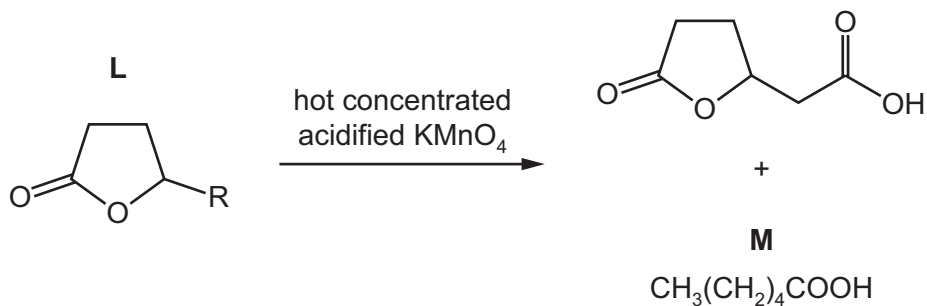
X

Z

[1]

Absorption Y shows that **L** has a C=C bond present in the R group.

The student decided to treat **L** with hot concentrated acidified potassium manganate(VII). The products of the reaction are shown.



(ii) Name **M**.

[1]

(iii) Use the information in (d) to deduce the molecular formula of **L**.

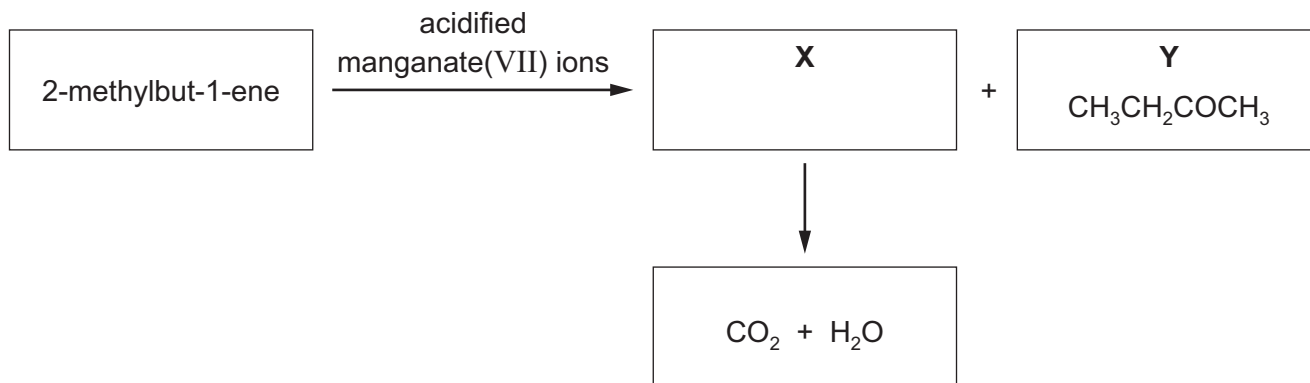
molecular formula of **L** =

[1]

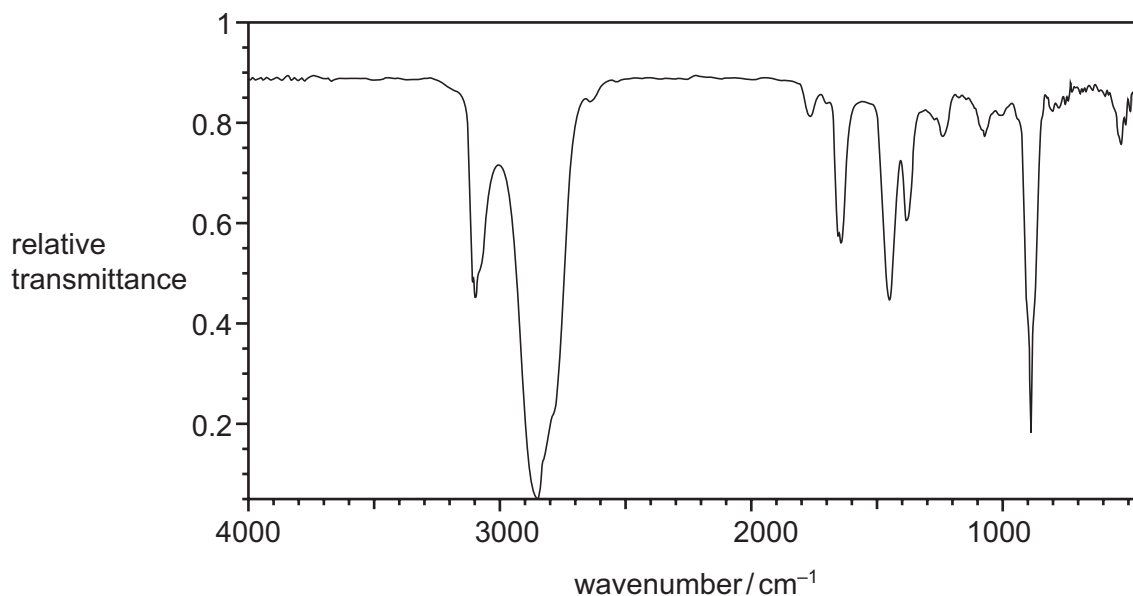
41. M/J 20/P21/Q6/e

- 6** 2-methylbut-1-ene reacts with acidified manganate(VII) ions, under specific conditions, to produce two organic compounds **X** and **Y**.

X immediately reacts with the acidified manganate(VII) ions to form carbon dioxide and water. **Y** has the structural formula $\text{CH}_3\text{CH}_2\text{COCH}_3$.



- (e)** The infra-red spectrum of 2-methylbut-1-ene is shown.



Predict two main differences that would be seen between the spectra of **Y**, $\text{CH}_3\text{CH}_2\text{COCH}_3$, and of 2-methylbut-1-ene. Give reasons for your predictions.

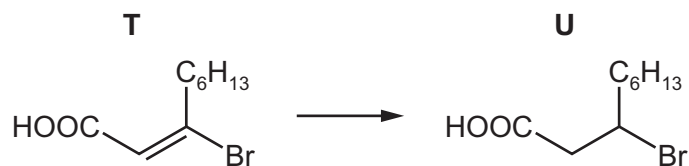
Your answer should refer only to the region of each spectrum **above** 1500 cm^{-1} .

[2]

42. MAR 20/P22/Q3/d(iii)

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

(d) A reaction of another unsaturated carboxylic acid, **T**, is shown.



(iii) The C–Br bond has an absorption between 500 cm^{-1} and 600 cm^{-1} in an infrared spectrum.

The infrared spectra for both **T** and **U** have absorptions between 2850 cm^{-1} and 2950 cm^{-1} . These correspond to C–H bonds.

Identify:

- two other absorptions that would be seen in the infrared spectra of both **T** and **U**
- one other absorption that would **only** be seen in the infrared spectrum of **T**.

For each absorption, give the range of the absorption and the bonds that correspond to these absorptions.

absorption 1 present in both spectra

absorption 2 present in both spectra

absorption **only** present in spectrum of **T**

[3]

43. O/N 19/P21/Q4/c(v)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm^{-1}
Y		red precipitate	1730 cm^{-1}
Z	no reaction	no reaction	$3200\text{--}3600\text{ cm}^{-1}$ 1630 cm^{-1} 1050 cm^{-1}

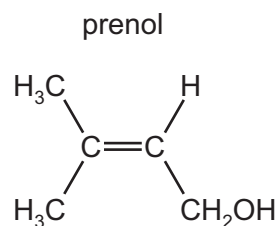
(v) Complete the table with the bond responsible for each of the principal absorptions seen in the infra-red spectrum of **Z**.

principal absorptions in infra-red spectrum	bond responsible
$3200\text{--}3600\text{ cm}^{-1}$	
1630 cm^{-1}	
1050 cm^{-1}	

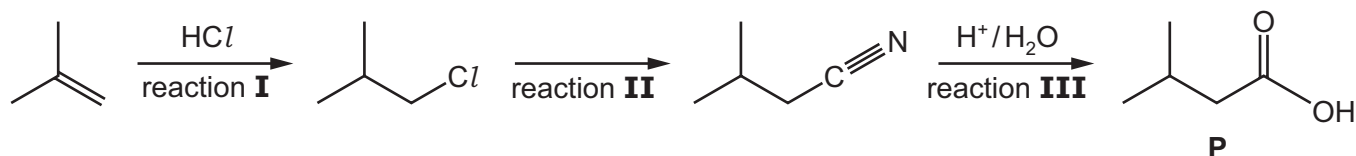
[1]

44. O/N 19/P22/Q4/e(i)

- 4** Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



(e) **P** can be produced as shown.



- (i)** The progress of reaction **I** can be monitored using infra-red spectroscopy.

One absorption that can be used to monitor the progress of this reaction is that of C–Cl at 730 cm^{-1} .

Identify another absorption that can be used to monitor the progress of this reaction. In your answer, you should refer to the specific bond and its corresponding absorption range in wavenumbers.

[1]

45. O/N 19/P23/Q4/c(v)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm^{-1}
Y		red precipitate	1730 cm^{-1}
Z	no reaction	no reaction	$3200\text{--}3600\text{ cm}^{-1}$ 1630 cm^{-1} 1050 cm^{-1}

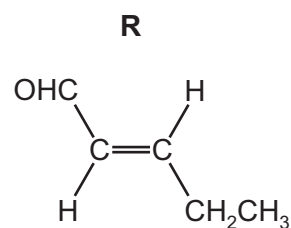
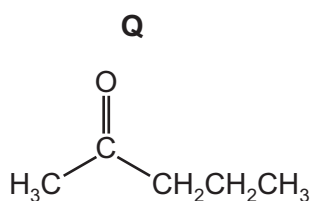
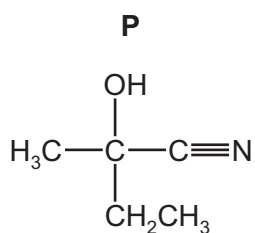
(v) Complete the table with the bond responsible for each of the principal absorptions seen in the infra-red spectrum of **Z**.

principal absorptions in infra-red spectrum	bond responsible
$3200\text{--}3600\text{ cm}^{-1}$	
1630 cm^{-1}	
1050 cm^{-1}	

[1]

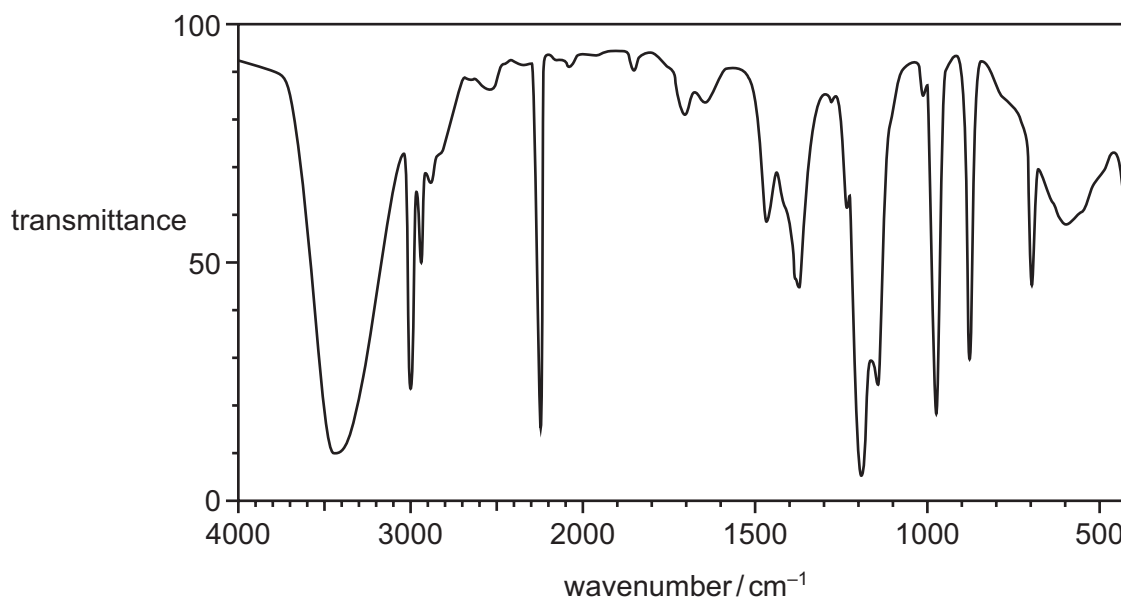
46. MAR 19/P22/Q3/d

3 **P**, **Q** and **R** all contain five carbon atoms.



A student carries out several tests to distinguish between **P**, **Q** and **R**.

(d) The infra-red spectrum shown corresponds to one of **P**, **Q** or **R**.



Deduce which of the compounds, **P**, **Q** or **R**, produces this spectrum. Explain your reasoning.

In your answer, identify any relevant absorptions in the infra-red spectrum and the bonds that correspond to these absorptions in the region **above** 1500 cm^{-1} .

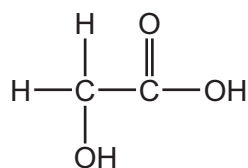
compound

explanation

[3]

47. O/N 18/P21/Q4/d(i)

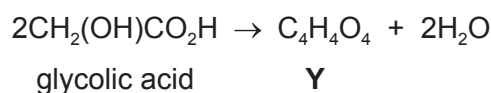
4 The structure of glycolic acid is shown.



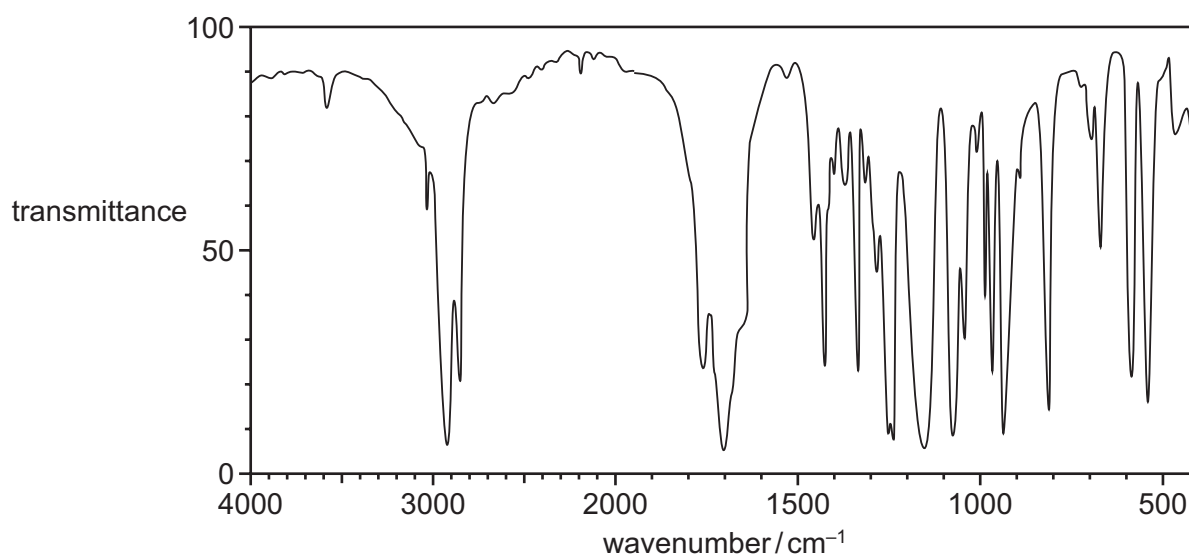
glycolic acid

(d) When glycolic acid is heated in the presence of a sulfuric acid catalyst, a new compound, **Y**, $\text{C}_4\text{H}_4\text{O}_4$, is formed.

The equation for the reaction is given.



(i) The infra-red spectrum of **Y** is shown.



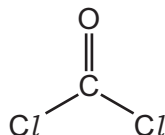
State how this spectrum differs from an infra-red spectrum of glycolic acid. Explain your answer with particular reference to the peaks within the range $1500\text{--}4000\text{ cm}^{-1}$.

[2]

48. O/N 18/P22/Q3/c(ii,iii)

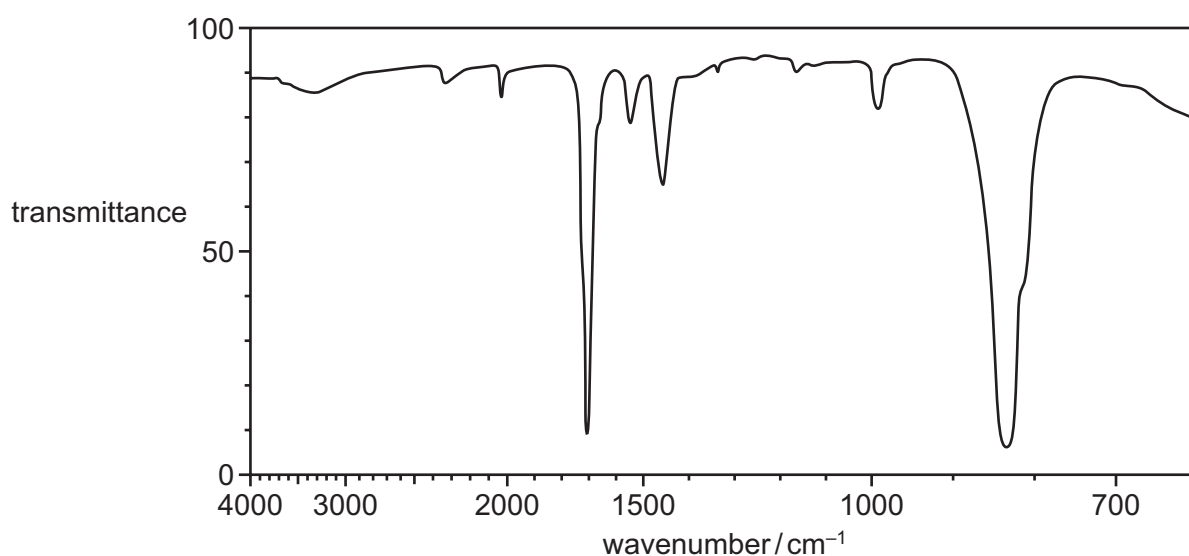
3 Trihalomethanes are organic molecules in which three of the hydrogen atoms of methane are replaced by halogen atoms, for example CHF_3 .

(c) A different trihalomethane, CHCl_3 , reacts with O_2 to produce carbonyl dichloride. HCl(g) is also released as a product of this reaction.



carbonyl dichloride

(ii) The conversion of CHCl_3 to carbonyl dichloride can be monitored by infra-red spectroscopy. The infra-red spectrum of carbonyl dichloride is shown.



On the infra-red spectrum of carbonyl dichloride identify with an **X** the absorption that would **not** be present in an infra-red spectrum of CHCl_3 .

Explain your answer.

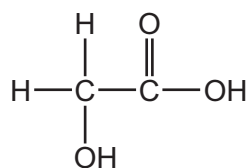
[2]

(iii) Suggest another difference between the infra-red spectra of CHCl_3 and carbonyl dichloride.

[1]

49. O/N 18/P23/Q4/d(i)

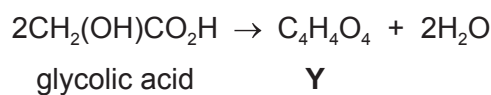
4 The structure of glycolic acid is shown.



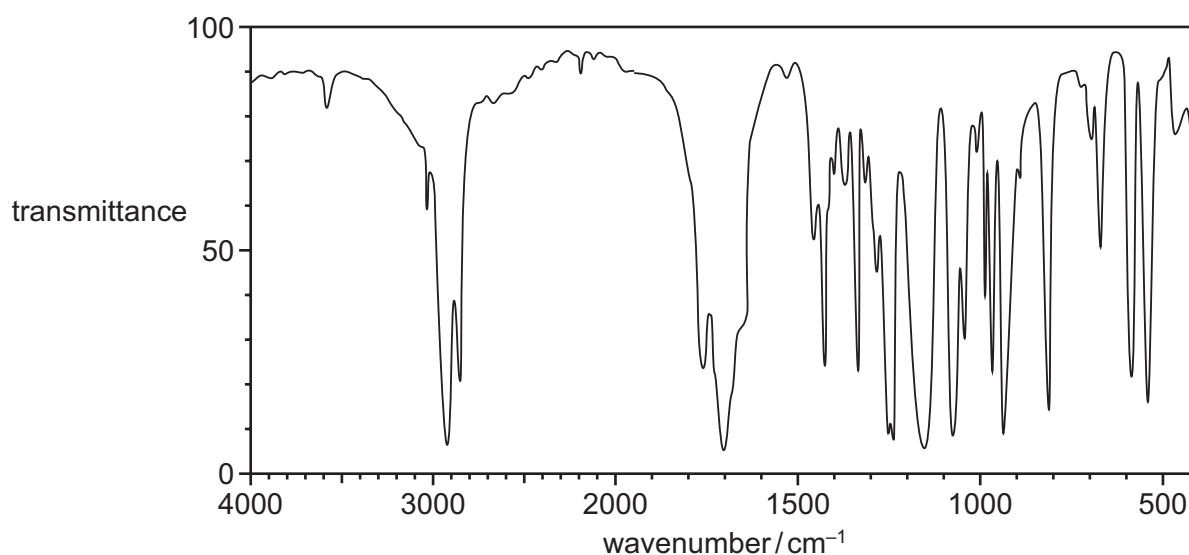
glycolic acid

(d) When glycolic acid is heated in the presence of a sulfuric acid catalyst, a new compound, **Y**, $\text{C}_4\text{H}_4\text{O}_4$, is formed.

The equation for the reaction is given.



(i) The infra-red spectrum of **Y** is shown.



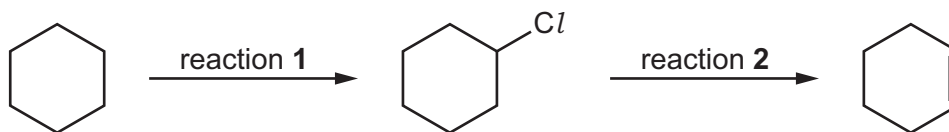
State how this spectrum differs from an infra-red spectrum of glycolic acid. Explain your answer with particular reference to the peaks within the range $1500\text{--}4000\text{ cm}^{-1}$.

[2]

50. MAR 18/P22/Q4/c(ii)

- 4** Cyclohexane is a colourless liquid used in industry to produce synthetic fibres.

A reaction scheme involving cyclohexane is shown.



- (c)** The product of reaction **2** is cyclohexene.

Cyclohexene can be converted into adipic acid (hexanedioic acid), $\text{HO}_2\text{C}(\text{CH}_2)_4\text{CO}_2\text{H}$.

- (ii)** Suggest **three** main differences between the infra-red spectra of cyclohexene and adipic acid.

In each case, identify the bond responsible and its characteristic absorption range (in wavenumbers).

1

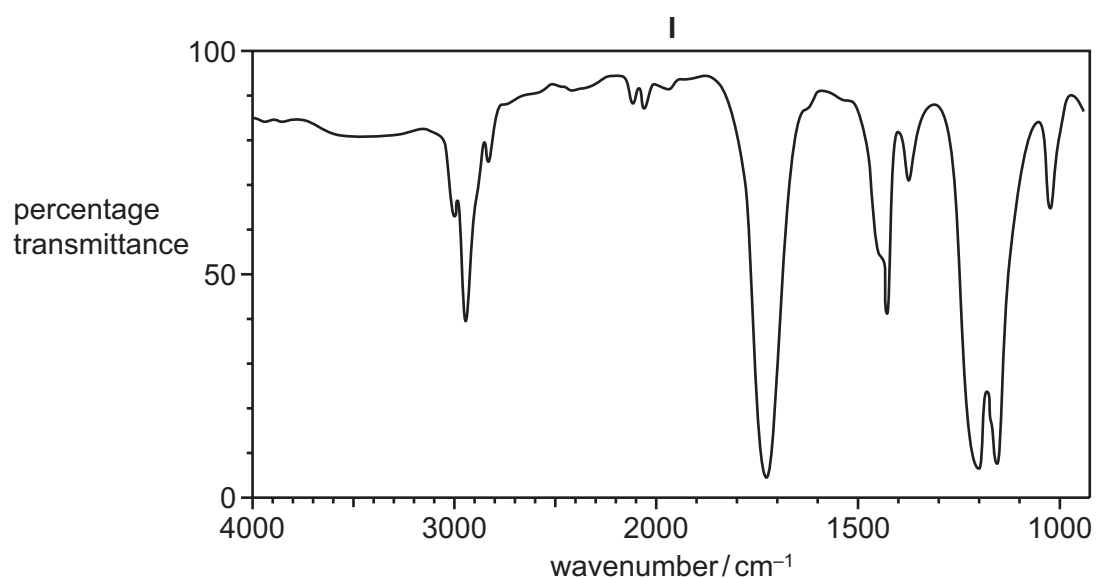
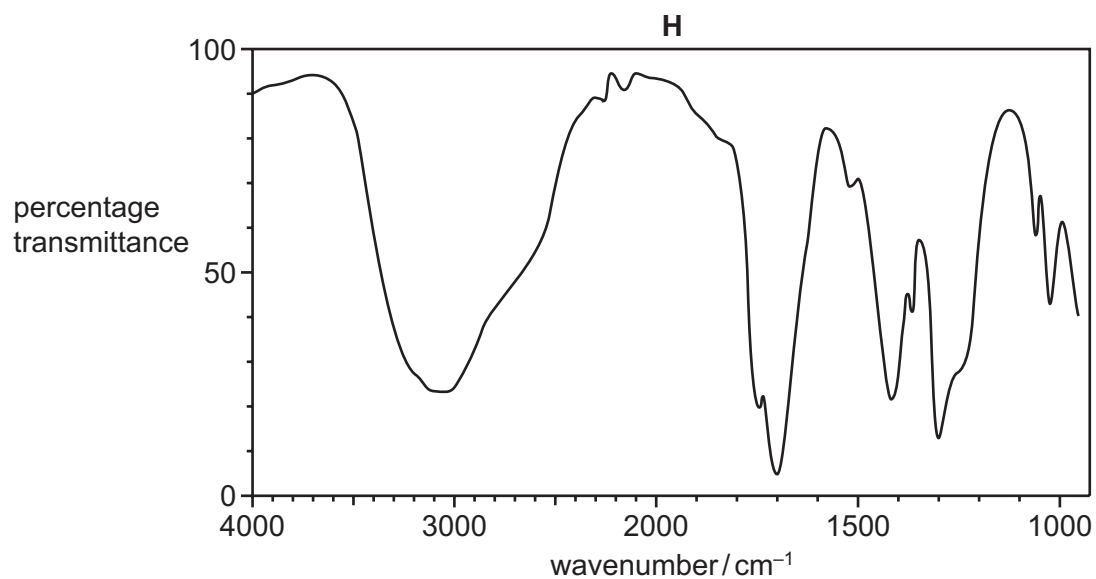
2

3

[3]

51. M/J 17/P22/Q4/d

(d) **H** and **I** are isomers with molecular formula $\text{C}_2\text{H}_4\text{O}_2$. The infra-red spectra of isomers **H** and **I** are shown.



(i) Identify the bonds responsible for the principal peaks above 1500 cm^{-1} in each spectrum.

spectrum of **H**

spectrum of **I**

[2]

(ii) Name **H** and **I**.

H

I

[2]

52. MAR 17/P22/Q3/d(ii)

(d) $\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{H}$ is a colourless liquid with an unpleasant odour.

It reacts with methanol in the presence of an acid catalyst to produce an organic product **V**, which has a pleasant fruity smell.

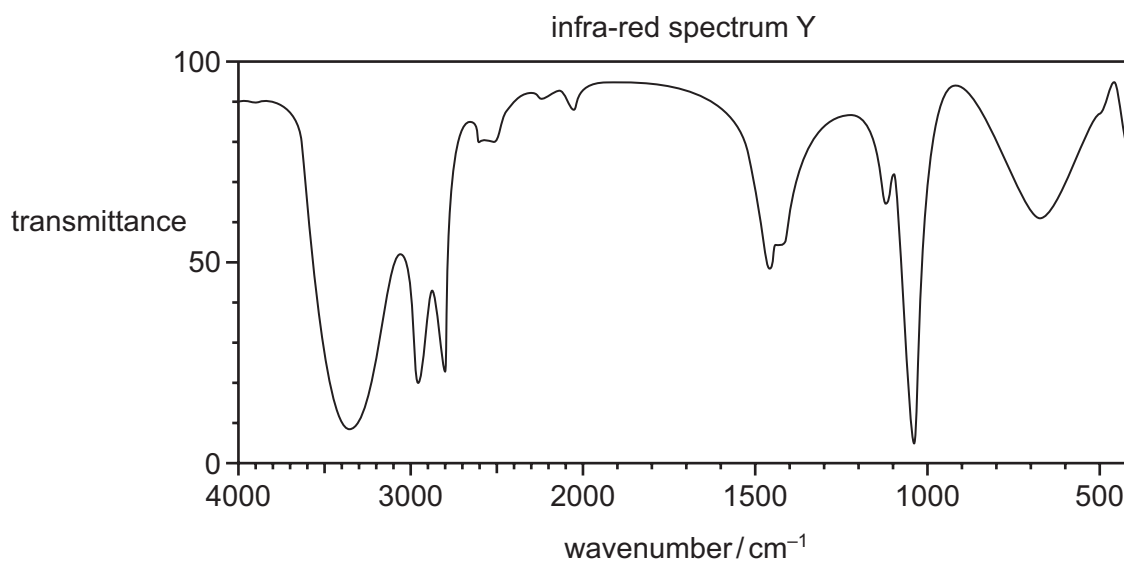
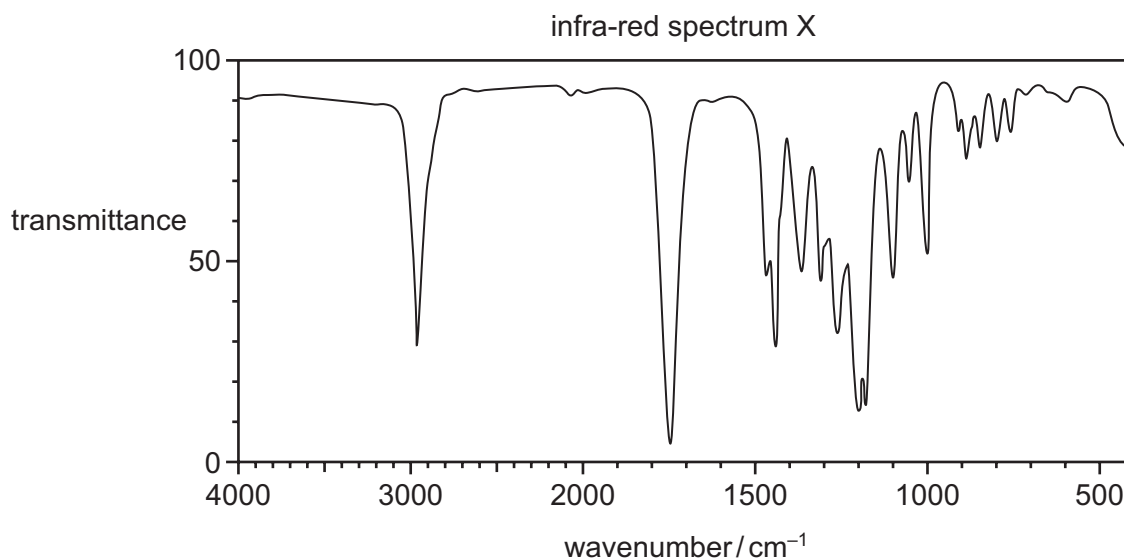
(ii) A student analysed $\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{H}$, methanol and **V** using infra-red spectroscopy. The spectra were returned to the student without labels.

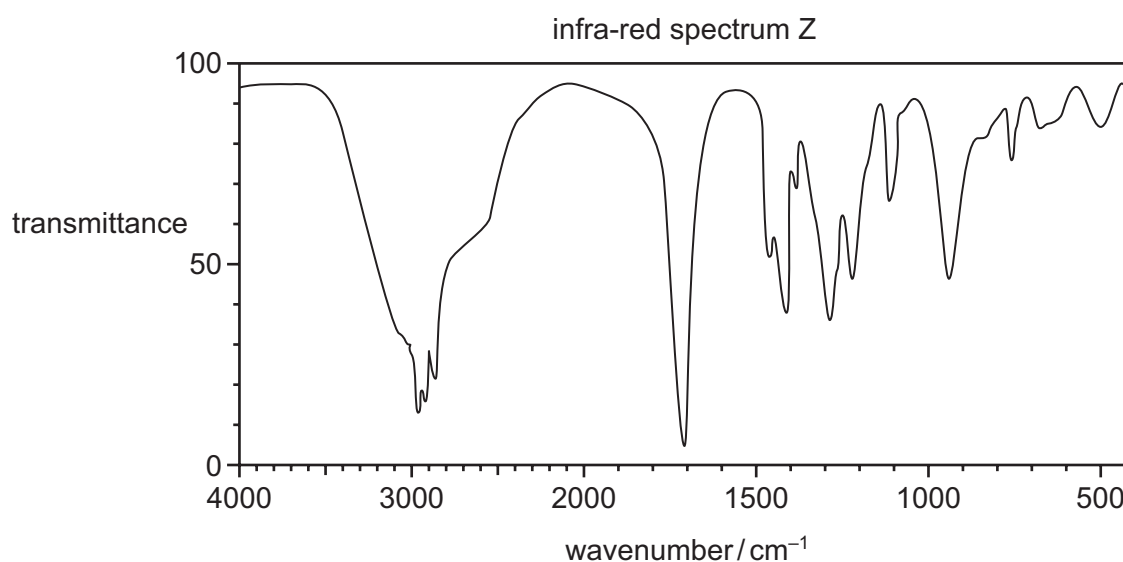
Identify which of the infra-red spectra, X, Y or Z, corresponds to **V**.

compound	$\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{H}$	methanol	V
spectrum			

Explain your answer with reference to relevant features of the **three** spectra in the region above 1500 cm^{-1} .

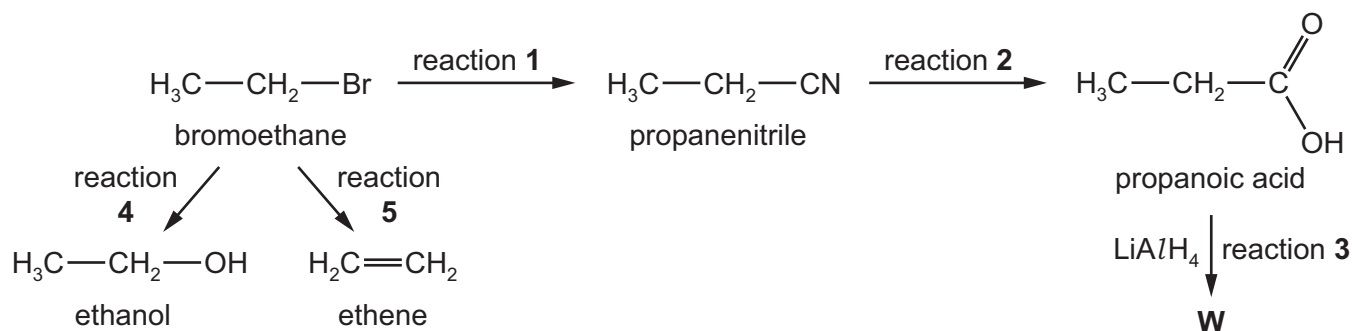
[4]





53. M/J 16/P21/Q5/b(ii)

5 A reaction sequence is shown.



(ii) Describe and explain the main difference between the infra-red spectrum of **W** and that of propanoic acid.

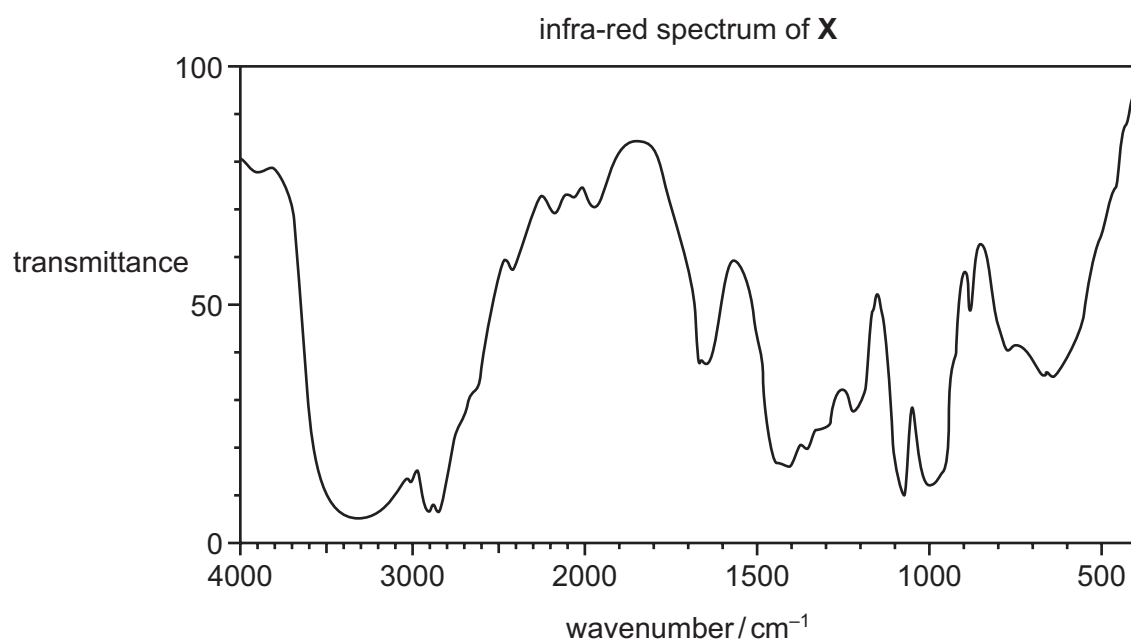
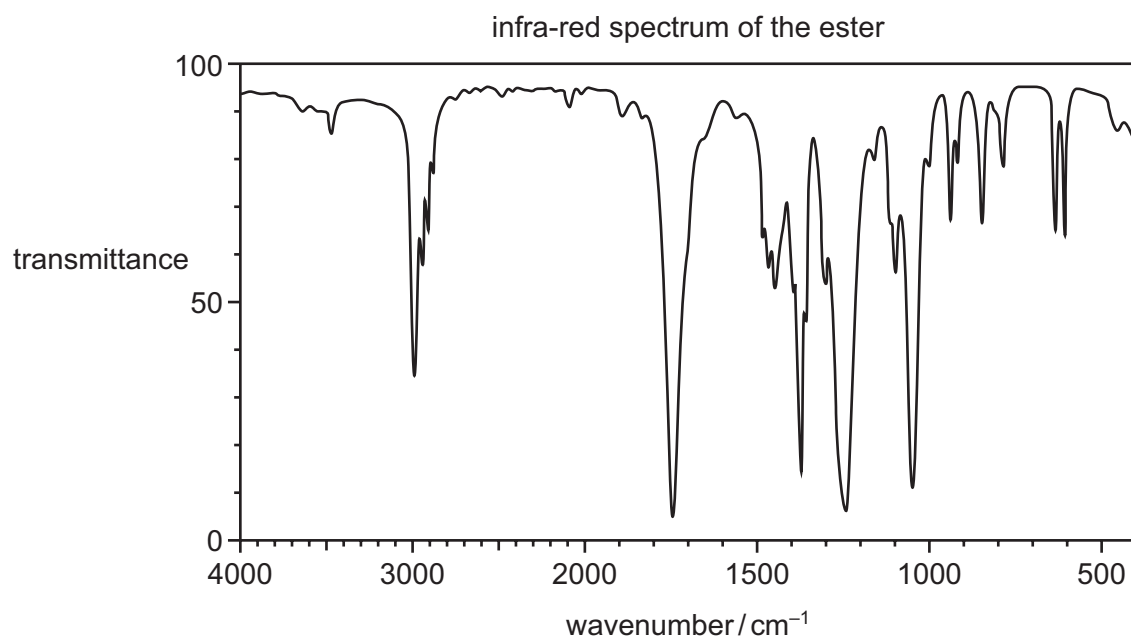
[2]

54. M/J 16/P22/Q4/c

4 This question is about molecules with molecular formula $C_4H_8O_2$.

(c) The infra-red spectra of one of the esters and of another isomer, **X**, are shown.

X decolourises bromine water and is not an ester or an acid.

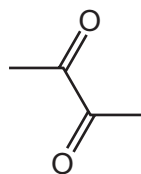
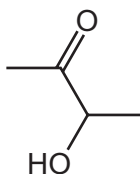


Explain the differences between these two spectra, with particular reference to the peaks with wavenumbers above 1500 cm^{-1} .

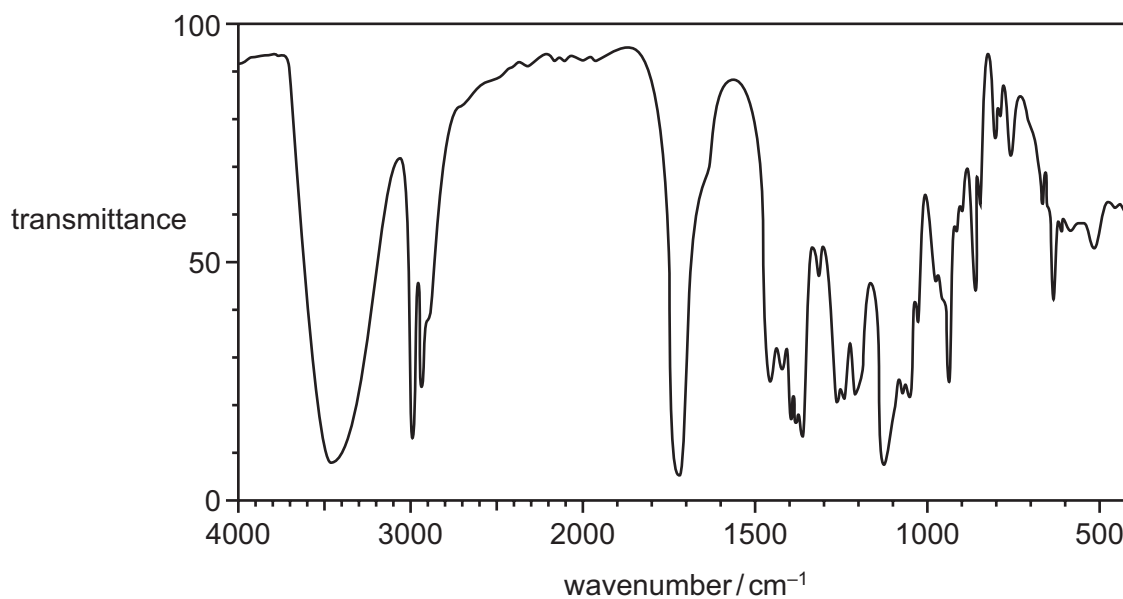
[3]

55. M/J 16/P23/Q4/c

- 4 Acetoin, $\text{CH}_3\text{COCH}(\text{OH})\text{CH}_3$, and diacetyl, $\text{CH}_3\text{COCOCH}_3$, are two of the compounds that give butter its characteristic flavour. Their skeletal formulae are shown.



- (c) The infra-red spectrum for acetoin is shown.



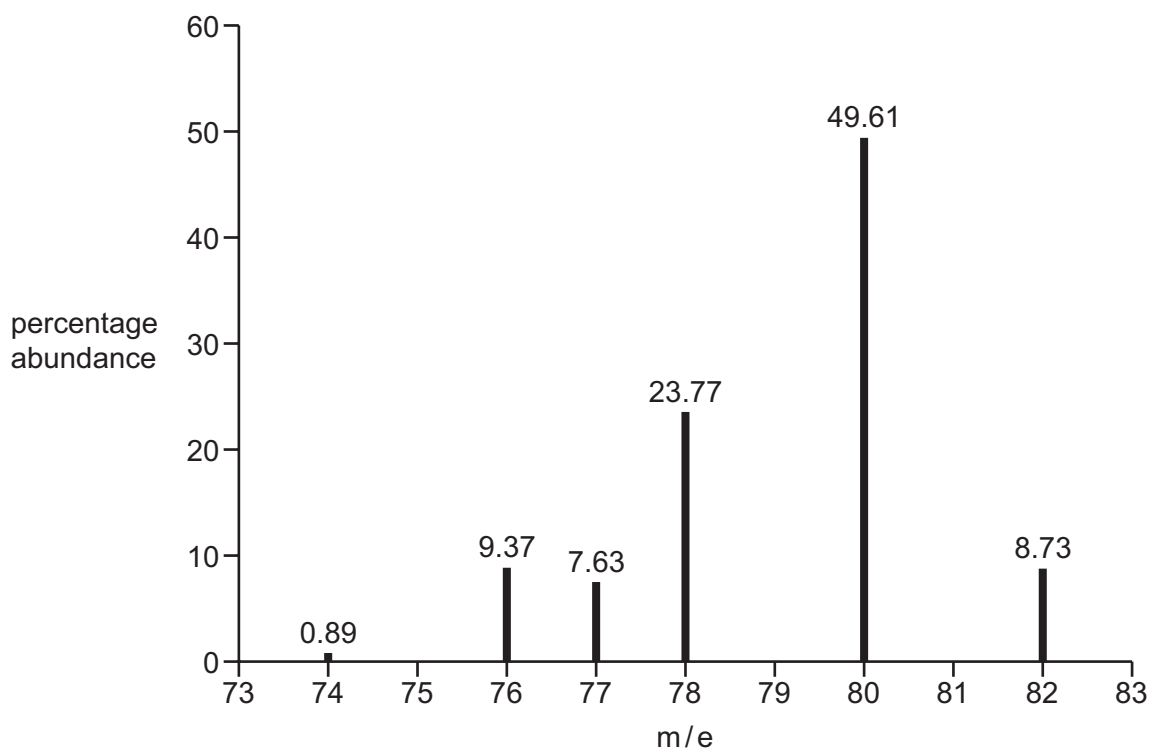
- (i) Explain the main features of this spectrum, with reference to the peaks with wavenumbers greater than 1500 cm^{-1} .
- [3]
- (ii) State and explain how the infra-red spectrum for diacetyl would differ from the infra-red spectrum for acetoin.

[2]

56. M/J 15/P21/Q1/b(ii)

(b) The relative atomic mass of an element can be determined using data from its mass spectrum.

The mass spectrum of element **X** is shown, with the percentage abundance of each isotope labelled.



(ii) Use the data in the mass spectrum to calculate the relative atomic mass, A_r , of **X**. Give your answer to **two** decimal places and suggest the identity of **X**.

A_r of **X**

identity of **X**

[2]

Answers Section

1. M/J 26/P21/Q5/g

5(g)(i) M1 $[\text{CH}_2\text{CH}^{37}\text{Cl}]^+$ M2 $[\text{CH}_2\text{CH}]^+$	2 M1 A $\text{C}_2\text{H}_3\text{Cl}^+$ where $\text{Cl} = \text{Cl-}^{37}$ A $\text{CH}_2\text{CHCl}^{37+}/[\text{CH}_2\text{CHCl}^{37}]^+$ A $\text{CH}_2=\text{CH}^{37}\text{Cl}^+$ I lack of brackets I dots M2 A C_2H_3^+ A $\text{CH}_2=\text{CH}^+$ A molecular formulas for both A any unambiguous formula for both A any unambiguous representation of ^{37}Cl Max 1 both without charges OR both negative charges OR both same incorrect positive charge (all else correct).
5(g)(ii) M1 2 chlorine isotopes (exist) M2 ^{35}Cl is 3 times more abundant than ^{37}Cl	2 75% ^{35}Cl and 25% ^{37}Cl ✓✓ ^{35}Cl is 3 times more abundant than ^{37}Cl ✓✓ ^{35}Cl is more abundant than ^{37}Cl ✓ M1 A m/e 64 is $[\text{M}+2]^+$ ion with ^{37}Cl M1 A peak at m/e 64 is due to ^{37}Cl AND peak at 62 is due to ^{35}Cl

2. M/J 26/P22/Q6/c(i,ii)

6(c)(i)	M1 (molecular ion peak at) 116 M2 has the same (116) molecular mass (as P)	2 M1 A the peak to be identified on the spectrum A the peak furthest to the right / the peak with the highest m/e value I an unquantified peak between 110 and 120 I 99 for HOOCCH=CHCO⁺ I 71 for C₃H₃O₂⁺ R any other peak except 99 and 71 M2 look at structures at the top of the page for evidence of the Mr of P = 116 I AMU M⁺ ion peak = molecular ion peak = M peak Molecular mass = M_r = RFM = molar mass A The molecular mass (of P) is 116 ✓✓ unless contradicted R relative atomic mass / A_r
6(c)(ii)	M1 (m / e = 45) COOH⁺ M2 (m / e = 99) HOOCCHCHCO⁺ / HOOCCH=CHCO⁺	2 Max 1 for both correct structures with no + charge OR – charge OR • OR no charges OR dots BOD sticks on fragment e.g. –COOH⁺ A CO₂H⁺ / CHO₂⁺ I C₄H₃O₃

3. M/J 26/P23/Q5/c

5(c)(i)	M1 two <i>Cl</i> isotopes M2 ^{35}Cl is 3 times more abundant than ^{37}Cl	2 M1 A (2) different <i>Cl</i> isotopes M1 BOD peaks are the isotopes of <i>Cl</i> 75% ^{35}Cl and 25% ^{37}Cl ✓✓ ^{35}Cl is 3 times more abundant than ^{37}Cl ✓✓ ^{35}Cl is more abundant than ^{37}Cl ✓^ $^{35}\text{Cl} : ^{37}\text{Cl}$ is 3 : 1 ✓✓
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5(c)(ii)	m / e value of peak	molecular formula of species responsible for peak	2	max 1 for (CH ₂ ³⁵ C _{l₂) (CH₂³⁵C^{β³⁷}C^l) (CH₂³⁷C_{l₂) ... i.e. all else correct but no charge OR negative charge OR shown as free radicals I lack of brackets I dots I position of positive charge}}
	84	(CH ₂ ³⁵ C _{l₂)⁺ BOD (C³⁷C^{β³⁵}C^l)⁺}		
	86	(CH ₂ ³⁵ C ^{β³⁷} C ^l) ⁺ BOD (C ³⁷ C _{l₂}) ⁺		
	88	(CH ₂ ³⁷ C _{l₂}) ⁺		

4. M/J 26/P24/Q4/c

4(c)	M1 bond: C=O (functional group: carboxyl / carbonyl) AND absorbance: 1670–1740 M2 bond: O–H AND absorbance: 2500–3500 AND functional group: carboxyl(ic acid)	2	M1 I references to C–H / C–O bonds A wavenumbers in the absorption range of 1670–1740 e.g. 1700 R functional group: amide / ester M2 A wavenumbers in the absorption range of 2500–3500 e.g. 2700 I OH A C=O group / O – H group R – OH and – O – H
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5. O/N 25/P21/Q3/e(ii)

3(e)(ii)	M1 absorption labelled at approx. 3000 cm ⁻¹ identified as O–H M2 absorption labelled at approx. 1720 cm ⁻¹ identified as C=O	2
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6. O/N 25/P21/Q4/c(ii,iii)

4(c)(ii)	(The spectrum is of) X AND 41 is [CH ₂ =CCH ₃] ⁺ / 41 is C ₃ H ₅ ⁺ OR 43 is [CH ₃ CHCH ₃] ⁺ / 43 is C ₃ H ₇ ⁺ OR X would NOT have significant peak at 54 / [(CH ₃) ₂ C=C ⁺] / C ₄ H ₆ ⁺ OR X would NOT have significant peak at 42 due to [(CH ₃) ₂ C ⁺] / C ₃ H ₆ ⁺	1
4(c)(iii)	([M+1] ⁺ / 34.7 × 100 / 1.1 = 6) [M+1] ⁺ = 2.29	1

7. O/N 25/P22/Q4/d(ii,iii)

4(d)(ii)	M1 absorption labelled at approx. 1750 cm ⁻¹ identified as C=O M2 absorption labelled at approx. 1650 cm ⁻¹ identified as C=C M3 absorption labelled at approx. 1170 cm ⁻¹ identified as C—O	3
4(d)(iii)	(0.31 / 4.7 × 100 / 1.1) = 6 carbon atoms	1

8. O/N 25/P23/Q3/e(ii)

3(e)(ii)	M1 absorption labelled at approx. 3000 cm ⁻¹ identified as O–H M2 absorption labelled at approx. 1720 cm ⁻¹ identified as C=O	2
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9. O/N 25/P23/Q4/c(ii,iii)

4(c)(ii)	(The spectrum is of) X AND 41 is $[\text{CH}_2=\text{CCH}_3]^+$ / 41 is C_3H_5^+ OR 43 is $[\text{CH}_3\text{CHCH}_3]^+$ / 43 is C_3H_7^+ OR X would NOT have significant peak at 54 / $[(\text{CH}_3)_2\text{C}=\text{C}^+]$ / C_4H_6^+ OR X would NOT have significant peak at 42 due to $[(\text{CH}_3)_2\text{C}^+]$ / C_3H_6^+	1
4(c)(iii)	$([\text{M}+1]^+ / 34.7 \times 100 / 1.1 = 6)$ $[\text{M}+1]^+ = 2.29$	1

10. M/J 25/P21/Q4/c

4(c)	Any two complete rows from: <table border="1"> <thead> <tr> <th>wavenumber</th><th>bond</th><th>functional group</th></tr> </thead> <tbody> <tr> <td>1040 – 1300</td><td>C–O</td><td>ester / hydroxy</td></tr> <tr> <td>3200 – 3650</td><td>O–H</td><td>hydroxy / alcohols</td></tr> <tr> <td>2850 – 2950</td><td>C–H</td><td>alkane</td></tr> <tr> <td>1710 – 1750</td><td>C=O</td><td>ester</td></tr> </tbody> </table>	wavenumber	bond	functional group	1040 – 1300	C–O	ester / hydroxy	3200 – 3650	O–H	hydroxy / alcohols	2850 – 2950	C–H	alkane	1710 – 1750	C=O	ester	2
wavenumber	bond	functional group															
1040 – 1300	C–O	ester / hydroxy															
3200 – 3650	O–H	hydroxy / alcohols															
2850 – 2950	C–H	alkane															
1710 – 1750	C=O	ester															

11. M/J 25/P22/Q5/b(iii)

5(b)(iii)	<table border="1"> <thead> <tr> <th>absorption / cm^{-1}</th><th>present in spectrum of vitamin C</th><th>present in spectrum of M</th></tr> </thead> <tbody> <tr> <td>1500–1680</td><td>✓</td><td>✗ / – / blank</td></tr> <tr> <td>2850–2950</td><td>✓</td><td>✓</td></tr> <tr> <td>3200–3650</td><td>✓</td><td>✓</td></tr> </tbody> </table>	absorption / cm^{-1}	present in spectrum of vitamin C	present in spectrum of M	1500–1680	✓	✗ / – / blank	2850–2950	✓	✓	3200–3650	✓	✓	1
absorption / cm^{-1}	present in spectrum of vitamin C	present in spectrum of M												
1500–1680	✓	✗ / – / blank												
2850–2950	✓	✓												
3200–3650	✓	✓												

12. M/J 25/P23/Q5/b(v)

5(b)(v)	A C–H alkane B C=O ester C C=C alkene D C–O ester 2 points = 1 mark, all 4 points = 2 marks	2
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13. M/J 25/P24/Q6/c

6(c)(i)	M1 $\frac{100}{1.1} \times \frac{11}{100}$ M2 10	2
6(c)(ii)	$\text{C}_{10}\text{H}_{18}$	1
6(c)(iii)	$(\text{C}_4\text{H}_9)^+$	1

14. MAR 25/P22/Q1/d

1(d)(i)	(relative) abundance is 3:1 / 100:33 AND link to two isotopes of <i>Cl</i>	1
1(d)(ii)	D is $\text{Cl}-\text{C}\equiv\text{C}-\text{H}$	1

15. O/N 24/P21/Q4/f

4(f)(i)	All three have a C—H OR CH bond	1
4(f)(ii)	compound A AND absorption at 2200–2250 cm ⁻¹ indicates C≡N	1

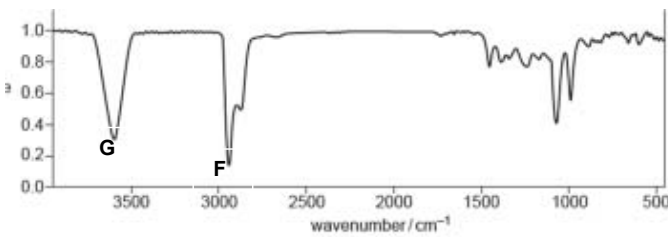
16. O/N 24/P23/Q4/f

4(f)(i)	All three have a C—H OR CH bond	1
4(f)(ii)	compound A AND absorption at 2200–2250 cm ⁻¹ indicates C≡N	1

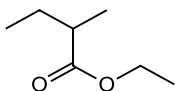
17. M/J 24/P21/Q6/c(i,iv)

6(c)(i)	M1 deduce $n = 12$ (from Y – 1C (in methanol)) M2 $(100 \times x) \div (1.1 \times 100) = 12$ (so) $x = 13.2$	2
6(c)(iv)	M1 Y will have absorption / peak / trough between 1670–1740 due to C=O (Q will not) M2 Q will have absorption / peak / trough between 3200–3600 due to O–H (Y will not)	2

18. M/J 24/P23/Q5/c

5(c)	 M1 peak shown around 2850–2950 labelled F AND (corresponds to) C—H M2 peak shown around 3200–3600 labelled G AND (corresponds to) O—H (in hydroxy(l) / alcohol)	2
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19. MAR 24/P22/Q4/c

4(c)(i)	$n = 100 / 1.1 \times 0.55 / 7.2$	1
	$= 6.94$ (so 7 carbon atoms)	1
4(c)(ii)	FG = ester / (R)COOR(')	1
	(absorption at) 1040–1300 cm ⁻¹ indicates C—O (stretch) AND (absorption at) 1710–1750 cm ⁻¹ indicates (C=O stretch)	1
4(c)(iv)	T 	1
	U 	1

20. O/N 23/P21/Q4/b

4(b)	$\frac{2.7}{48.7} \times \frac{100}{1.1} = 5(.04)$	1
	$M_r = 5 \times 12 + 10 \times 1 = 70$	1

21. O/N 23/P22/Q3/c(iv),d(ii)

3(c)(iv)	C=O	1
	absorption changes to frequency / wavenumber / from 1670–1740 to 1710–1750 cm ⁻¹	1
3(d)(ii)	$\frac{0.84}{12.7} \times \frac{100}{1.1} (= 6.013)$	1
	no. of carbon atoms = 6	1
	molecular formula = C ₆ H ₁₅ O ₄ P	1

22. O/N 23/P23/Q4/b

4(b)	$\frac{2.7}{48.7} \times \frac{100}{1.1} = 5(.04)$	1
	$M_r = 5 \times 12 + 10 \times 1 = 70$	1

23. M/J 23/P21/Q3/f

3(f)	M1 <i>m/e</i> 164 (CF ₂ ³⁵ Cl ⁷⁹ Br) ⁺ M2 <i>m/e</i> 166 (CF ₂ ³⁷ Cl ⁷⁹ Br) ⁺ AND (CF ₂ ³⁵ Cl ⁸¹ Br) ⁺	2
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24. M/J 23/P22/Q5/d(iv)

5(d)(iv)	Any two of the following differences in spectra D compared to E :			2
		bond	functional group	
	E	C=O / C double bond to O	(ester)	
	D	C=O	(carboxy(l))	
	D	O-H OR OH bond	(carboxy(l))	
	D	O-H OR OH bond	(hydroxy(l) OR alcohol)	

25. M/J 23/P23/Q1/c

1(c)	⁶³ Cu ²⁺	1
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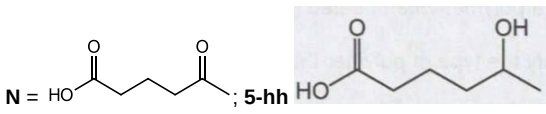
26. M/J 23/P23/Q6/b(iv)

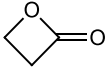
6(b)(iv)	S C≡N AND T C=C	1
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27. MAR 23/P22/Q4/d

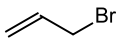
4(d)(i)	G	1
	(absorption at) 2200–2250 (cm ⁻¹) AND C≡N	1
4(d)(ii)	3.4 AND relative abundance of ⁷⁹ Br: ⁸¹ Br ≅ 50:50 OR 1:1	1

28. O/N 22/P21/Q5/b,c

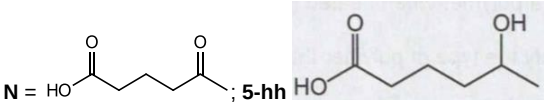
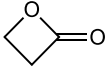
5(b)	 N = HO-C(=O)-CH₂-CH₂-C(=O)-CH₃; 5-hh HO-C(=O)-CH₂-CH₂-CH₂-CH(OH)-CH₃	1
	M1 absorptions will overlap / be similar / the same / indistinguishable	
	M2 both have some bonds in similar environments owtte	1

5(c)	$100 / 1.1 \div 95.5 / 3.15 = 3$ carbon atoms	1
		1

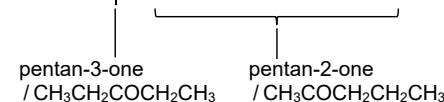
29. O/N 22/P22/Q3/c

3(c)(i)	C=C (bond) / alkene AND (absorption within) $1500\text{--}1680\text{ cm}^{-1}$ (present in E's IR spectrum only)	1
3(c)(ii)	hydroxyl / alcohol AND (broad absorption within) $3200\text{--}3650\text{ cm}^{-1}$ (present in F's IR spectrum only)	1
3(c)(iii)	120.9 / 121	1
3(c)(iv)		1
3(c)(v)	$(1)\text{H}_2\text{O} + (1)[\text{O}] + \text{C}_3\text{H}_5\text{Br} \rightarrow \text{C}_3\text{H}_7\text{BrO}_2$	1

30. O/N 22/P23/Q5/b,c

5(b)	 ----- M1 absorptions will overlap / be similar / the same / indistinguishable M2 both have some bonds in similar environments owtte	1
5(c)	$100 / 1.1 \div 95.5 / 3.15 = 3$ carbon atoms	1
		1

31. M/J 22/P21/Q4/d

4(d)	29 C_2H_5^+ 57 $\text{COCH}_2\text{CH}_3^+ \text{ OR } \text{C}_3\text{H}_5\text{O}^+ \text{ OR } \text{CH}_2\text{COCH}_3^+$ identity of Z 	3
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32. M/J 22/P22/Q6/c(i,ii)

6(c)(i)	Look for reference to 116 as the mass of the molecular ion AND mass of $\text{C}_{(1)}\text{H}_{(1)}\text{O}_{(1)} = 29$ to conclude molecular formula is $\text{C}_4\text{H}_4\text{O}_4$ $116 / 29 = 4$ so $\text{C}_4\text{H}_4\text{O}_4$	1
6(c)(ii)	M1 m/e 45: $^+\text{COOH}$ OR $^+\text{CHO}_2$ M2 m/e 71: $\text{C}_3\text{H}_3\text{O}_2^+$	2

33. M/J 22/P23/Q5/b

5(b)(i)	$100 \times 2.2 / 1.1 \times 100 = 2$	1
5(b)(ii)	chlorine / Cl AND peak at M+2 represents the molecular ion with 37-Cl (rather than 35-Cl as relative abundance of (peaks) M: M+2 is $100:33 / 3:1$) OR relative abundance of (peaks) M: M+2 is $100:33.3 / 3:1$ (so peak at M+2 contains 37-Cl)	1

5(b)(iii)	m / e = 29: C ₂ H ₅ ⁺ m / e = 49: CH ₂ C ⁺ name of Z: chloroethane	3
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34. MAR 22/P22/Q4/b(iii)

4(b)(iii)	Perspex® would not have absorption 1500–1680 cm ⁻¹ AND Perspex® does not have C=C	1
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35. O/N 21/P21/Q4/b

4(b)	M1 ester M2 1100 cm ⁻¹ linked to C—O AND 1720 cm ⁻¹ linked to C=O M3 No COOH / carboxylic acid and No OH / alcohol in D (but present in C) OR COOH / carboxylic acid and OH / alcohol reacted /lost (in C to form D)	3
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36. O/N 21/P22/Q3/d(iii)

3(d)(iii)	M1 C—O in range 1040–1300 (cm ⁻¹) M2 C=O in range 1670–1740 (cm ⁻¹)	2
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37. O/N 21/P23/Q4/b

4(b)	M1 ester M2 1100 cm ⁻¹ linked to C—O AND 1720 cm ⁻¹ linked to C=O M3 No COOH / carboxylic acid and No OH / alcohol in D (but present in C) OR COOH / carboxylic acid and OH / alcohol reacted /lost (in C to form D)	3
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38. M/J 21/P22/Q5/b(v)

5(b)(v)	2200–2250 (cm ⁻¹ due to) C≡N / triple bond between C and N.	1
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39. MAR 21/P22/Q3/c

3(c)	M1: 2150–2250 (cm ⁻¹) AND C≡C M2: 2200–2250 (cm ⁻¹) AND C≡N	2
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40. O/N 20/P22/Q4/d

4(d)(i)	X is C=O AND Z is C—O	1
4(d)(ii)	hexanoic acid	1
4(d)(iii)	C ₁₂ H ₂₀ O ₂	1

41. M/J 20/P21/Q6/e

6(e)	<i>Predict two differences, with reasons, between spectra of Y, CH₃CH₂COCH₃ and 2-methylbut-1-ene (shown)</i> <i>first difference</i> M1 absence of peak/ absorption at 3100 (cm ⁻¹) as no longer any =C—H present (in Y) <i>second difference</i> M2 peak at 1650 (cm ⁻¹) moves to the left to any value / range of values between 1670 and 1740) due to disappearance of C=C (in Y) and appearance of C=O (in Y) OR absence of peak at 1650 (cm ⁻¹) as no longer any C=C present (in Y) AND appearance of peak (in Y) at (any value / range of values) between 1670-1740(cm ⁻¹) due to C=O	2
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42. MAR 20/P22/Q3/d(iii)

3(d)(iii)	M1 / M2 absorptions seen in both spectra (any two): (same) both show an absorption at 1680–1730 (cm^{-1}) because of C=O (same) both show an absorption at 1040–1300 (cm^{-1}) because of C–O (same) both show an absorption at 2500–3000 (cm^{-1}) because of RCO_2H / O–H in RCO_2H / carboxyl(ic acid) M3 absorption only seen in spectrum of T: (different) T shows an absorption at 1500–1680 (cm^{-1}) because of C=C (different) T shows an absorption at 3000–3100 (cm^{-1}) because of (C=)C–H	3
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43. O/N 19/P21/Q4/c(v)

4(c)(v)	principal absorptions in the infra-red spectrum	bond responsible	1
	3200–3600 cm^{-1}	RO–H / O–H	
	1630 cm^{-1}	C=C	
	1050 cm^{-1}	C—O	

44. O/N 19/P22/Q4/e(i)

4(e)(i)	1500–1680 (cm^{-1}) AND C=C	1
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45. O/N 19/P23/Q4/c(v)

4(c)(v)	principal absorptions in the infra-red spectrum	bond responsible	1
	3200–3600 cm^{-1}	RO–H / O–H	
	1630 cm^{-1}	C=C	
	1050 cm^{-1}	C—O	

46. MAR 19/P22/Q3/d

3(d)	M1 compound P M2 (absorption at) 2250 cm^{-1} AND C≡N (stretch) M3 (absorption at) 3100–3700 cm^{-1} AND O—H (stretch)	3
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47. O/N 18/P21/Q4/d(i)

4(d)(i)	EITHER Glycolic acid would have: M1 2500–3000 due to RCO_2H M2 range within 3200–3650 due to RO–H OR Spectrum Y would NOT have: M1 2500–3000 due to RCO_2H M2 range within 3200–3650 due to RO–H	2
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48. O/N 18/P22/Q3/c(ii,iii)

3(c)(ii)	M1 X marked on peak at 1670–1740 cm^{-1} [1] M2 CHCl_3 has no C=O [1]	2
3(c)(iii)	(It / CHCl_3 has a) peak at 2850–2950 (cm^{-1}) OR carbonyl dichloride spectrum has no peak 2850–2950 (cm^{-1})	1

49. O/N 18/P23/Q4/d(i)

4(d)(i)	EITHER Glycolic acid would have: M1 2500–3000 due to RCO_2H M2 range within 3200–3650 due to ROH OR Spectrum Y would NOT have: M1 2500–3000 due to RCO_2H M2 range within 3200–3650 due to ROH	2
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50. MAR 18/P22/Q4/c(ii)

4(c)(ii)	cyclohexene would have absorption at 1500–1680 (cm^{-1}) because of $\text{C}=\text{C}$ (and adipic acid would not) cyclohexene would have absorption at 3000–3100 (cm^{-1}) because of $\text{C}=\text{H}/\text{C}-\text{H}$ in alkene (and adipic acid would not) adipic acid would have absorption at 2500–3000 (cm^{-1}) because of $\text{O}-\text{H}/\text{CO}_2-\text{H}$ (and cyclohexene would not) adipic acid would have absorption at 1040–1300 (cm^{-1}) because of $\text{C}-\text{O}$ (and cyclohexene would not) adipic acid would have absorption at 1640–1750 (cm^{-1}) because of $\text{C}=\text{O}$ (and cyclohexene would not)	max 3
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51. M/J 17/P22/Q4/d

4(d)(i)	H $\text{C}=\text{O}$ (group / bond) AND $\text{O}-\text{H}$ (group / bond)	1
	I $\text{C}=\text{O}$ (group / bond) AND $\text{C}-\text{H}$ (group / bond)	1
4(d)(ii)	H = ethanoic acid	1
	I = methyl methanoate	1

52. MAR 17/P22/Q3/d(ii)

3(d)(ii)	(compound V is) spectrum X	1
	spectra X and Z show a $\text{C}=\text{O}$ (stretch) at 1730 (cm^{-1})	1
	spectra Y and Z show $\text{O}-\text{H}$ (stretches) above 2500 (cm^{-1})	1
	V has a $\text{C}=\text{O}$ (bond) and no $\text{O}-\text{H}$ (bond)	1

53. M/J 16/P21/Q5/b(ii)

(ii)	disappearance of peak / dip / trough / absorption at 1680–1730	[1]	[2]
	due to (loss of) $\text{C}=\text{O}$	[1]	
	OR		
	peak at 3200–3650	[1]	
	due to (alcohol) $\text{O}-\text{H}$ (formation)	[1]	

54. M/J 16/P22/Q4/c

(c)	Peak at 1710–1750 (for ester) due to $\text{C}(=\text{O})$ Peak at 1500–1680 (for X) due to $\text{C}(=\text{C})/\text{alkene}$ Peak at 3200–3650 (for X) due to (alcohol) $\text{O}-\text{H}$	[1] [1] [1]	[3]
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55. M/J 16/P23/Q4/c

(c) (i)	absorption at 1670–1740 $\text{C}(=\text{O})$ absorption at 2850–3000 $\text{C}-\text{H}$ absorption at 3200–3650 $\text{O}-\text{H}$	[1] [1] [1]	[3]
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(ii)	no absorption at 3200–3650 O-H disappears / no O-H bond in diacetyl	[1] [1]	[2]
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56. M/J 15/P21/Q1/b(ii)

(ii)	$\frac{(0.89 \times 74) + (9.37 \times 76) + (7.63 \times 77) + (23.77 \times 78) + (49.61 \times 80) + (8.73 \times 82)}{100}$ = 79.04 (2 d.p.) AND Se	[1] [1]	 [2]
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